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The Evolution of Interest-Rate Models: From the Yield Curve to the Swaption Cube

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ABSTRACT

Interest-rate modelling is often taught as a catalogue of competing stochastic equations, obscuring why models were created and why modern sell-side desks use several simultaneously. This survey reorganises the field around five layers of a pricing architecture: curve construction; arbitrage-free dynamics; volatility-smile representation; collateral, multi-curve and risk-free-rate conventions; and numerical implementation. Within this framework, the progression from Vasicek and Cox–Ingersoll–Ross to Hull–White, HJM, the LIBOR market model, SABR and post-LIBOR forward market models is interpreted as a sequence of responses to new products, quotation practices, computational constraints and market institutions—not as a tournament won by one equation. We derive the core equations, compare the principal model families, connect calibration choices to hedging instruments, and map models to major rates products and XVA applications. Particular attention is paid to risks concealed by exact vanilla calibration, including correlation, conditional curve dynamics, future smile behavior and numerical error. An open Python/C++ companion repository provides reproducible illustrations for selected stages. The resulting framework offers a systematic introduction for new quantitative analysts and a practical reference for researchers, traders, model developers and validators.

“An interest-rate model is not merely an equation for rates; it is a disciplined account of which market quantities must agree today, how they may evolve tomorrow, and which risks remain when calibration is complete.”

1 | Introduction: An Interest-Rate Model is a Pricing Architecture, not a Single Equation

Few areas of quantitative finance contain so many famous models and so little agreement about what the word *model* denotes. A macroeconomist may use an interest-rate model to describe

expectations, term premia, and monetary-policy transmission. A bond investor may want a parsimonious representation of level, slope, and curvature. A derivatives trader needs prices and hedge ratios consistent with today's curve and option market. A model-validation team asks whether the implementation is stable outside calibration instruments. An XVA engine needs joint future scenarios under collateral and netting conventions. These are related problems, but they do not have the same state variables, probability measure, loss function, or standard of success.

The distinction is especially sharp between buy-side and sell-side modelling. Under the physical measure, an investor cares about forecasts and risk premia; under a pricing measure, a dealer primarily cares about replication, calibration,

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interpolation across liquid quotes, and the behavior of hedges. Nawalkha and Rebonato (2011) make this distinction explicit. A model that forecasts yields well can be unsuitable for pricing a Bermudan swaption, while a model that reproduces every liquid swaption may say little about expected bond returns. This tension is also illustrated in Section 8, which turns briefly to the complementary literature that validates term-structure models against historical yield dynamics and against the joint pricing of interest-rate options, rather than against today's calibration instruments alone. Sell-side models usually start elsewhere: today's market is taken as given, and the model specifies arbitrage-free evolution sufficiently rich to value claims that cannot be replicated statically.

This survey therefore does not rank models on a single scale. It proposes a five-layer architecture:

1. **Curve Representation:** What discount factors, forward rates, basis curves, and collateral curves are implied by traded cash instruments?
2. **Dynamic State:** Which quantities move—short rates, instantaneous forwards, discrete forwards, swap rates, or latent factors—and under which numeraire?
3. **Smile Representation:** How are strike dependence, volatility-of-volatility, rate-level dependence, and negative rates represented?
4. **Market Infrastructure:** Which benchmark, collateral, funding, settlement, and compounding conventions define the payoff and discounting?
5. **Numerical Realization:** Which analytic approximation, lattice, partial differential equation (PDE), Monte Carlo scheme, calibration routine, and sensitivity method makes the model operational?

Table 1 is the organizing device for the paper. A model belongs primarily to one layer, although it can span several. Nelson–Siegel is mainly a curve-shape representation; Hull–White is mainly a low-dimensional dynamic model; SABR is mainly a smile model; the LIBOR market model (LMM) is a joint forward-rate dynamic; and overnight-index discounting is market infrastructure. Calling these direct substitutes produces category errors. Nelson–Siegel does not replace SABR, and SABR does not by itself generate an arbitrage-consistent future yield curve. Likewise, the Black formula is not a full term-structure model: it is a quotation and pricing convention for a single European option under an appropriate measure.

The historical development in Figure 1 can be read as a succession of modelling bottlenecks. Bachelier (1900) and Black (1976) supplied European-option languages. Vasicek (1977) and Cox et al. (1985) supplied tractable short-rate economies. Ho and Lee (1986), Black et al. (1990), Hull and White (1990), and Black and Karasinski (1991) made exact or practical fitting of the observed term structure central.

Prior to Duffie and Kan (1996), most term-structure models like Vasicek (1977) and Cox et al. 1985 existed as isolated single/multi-factor models with explicit pricing solutions. Duffie and Kan (1996) formalized and unified the class of Affine Term Structure

EVOLUTION OF INTEREST-RATE MODELLING

From marginal option laws to integrated pricing architectures

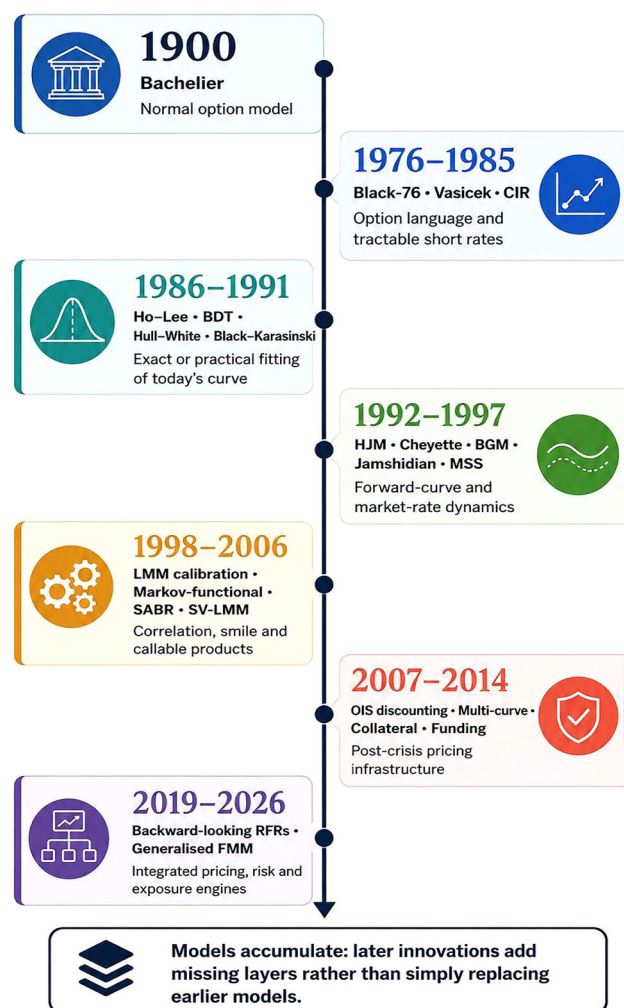


FIGURE 1 | The intellectual and market evolution of interest-rate modelling. Dates identify publication or market-adoption landmarks rather than clean regime breaks. Old models survive when they remain efficient for a particular product, while later models usually add a missing layer rather than replacing every earlier layer. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

Models (ATSMs) in continuous time as well as provided a general, rigorous framework that established the exact mathematical conditions under which zero-coupon bond prices are exponential-affine functions of state variables. Building on the foundation laid by Duffie and Kan (1996), Dai and Singleton (2000) provided the definitive taxonomy $A_m(N)$, admissibility theory, and identification framework, which transformed ATSMs from a collection of isolated equations into a systematically classified, econometrically estimable mathematical family.

Heath et al. (1992) moved the object of modelling to the forward curve and derived drift from no arbitrage. Brace et al. (1997), Jamshidian (1997), and Miltersen et al. (1997) then modelled market-quoted forward rates directly. Hagan et al. (2002) supplied a compact stochastic-volatility language for smile. The global financial crisis forced discounting and forwarding apart, and the LIBOR transition later required models of backward-looking

TABLE 1 | The five layers of a sell-side interest-rate modelling system.

Layer	Primary object	Representative methods	Main observable inputs	Typical output	Principal failure mode
1. Curve	$P^d(t, T)$, projection forwards $F^x(t; T, T_{t+1})$	bootstrap, splines, monotone-convex interpolation, Nelson–Siegel–Svensson	deposits, futures, OIS, swaps, basis swaps	discount factors, zero rates, forwards, bucketed curve risk	exact repricing but unstable or nonlocal risk
2. Dynamics	$r_t, f(t, T), L_t(t), S_{\alpha, \beta}(t)$	Vasicek, CIR, Hull–White, G2++, HJM, Cheyette, LMM, Markov-functional	initial curves and selected option prices	future paths and conditional distributions	plausible vanillas but wrong joint/conditional dynamics
3. Smile	$\sigma_{\text{imp}}(T, K)$ and its dynamics	Black, Bachelier, displaced diffusion, local volatility, SABR, stochastic-volatility LMM	caplet and swaption volatility cube	strike interpolation, smile Greeks, tail prices	static fit with unstable or unrealistic smile moves
4. Infrastructure	collateral account, benchmark and tenor basis	OIS discounting, multi-curve framework, RFR compounding, fallback spread	CSA terms, OIS, basis, RFR futures/swaps	collateralised values and benchmark-consistent cash flows	internally elegant model of the wrong contract
5. Numerics	discretised pricing and risk engine	closed form, tree, PDE, Monte Carlo, LSM, AAD	calibrated parameters and trade definition	PV, Greeks, P&L explain, exposure	discretisation, regression, optimisation, or implementation error

compounded overnight rates (Mercurio 2009; Piterbarg 2010; Lyashenko and Mercurio 2019, 2020).

The institutional pull on model development was strongest where the option book was largest. In the United States agency mortgage market, the securitised prepayment option made behavioural early exercise an industrial-scale modelling problem decades before the swaption cube existed (see Section 6.5).

The paper makes four contributions relative to existing surveys. First, it focuses on the economic and institutional logic of sell-side model evolution. Rebonato (2004a, 2023) provides exceptionally clear accounts of term-structure and forward-rate modelling; Gibson et al. (2010) survey the wider term-structure literature. The present survey expands the unit of analysis from a stochastic process to the entire pricing stack. Second, it derives the minimum mathematical core needed to move coherently from discount factors to numeraires, short-rate models, HJM, LMM, smile models, and multi-curve valuation. Third, it offers a product-to-model map and a validation hierarchy, emphasising the risks that are not identified by vanilla calibration. Fourth, it links theory to reproducible implementation. Selected constructions are illustrated in Feng's (2026) open [Rates-Pricing-Quant-Library](#)¹, which provides notebook-based Python and C++ examples for curve construction, parametric curve fitting, swaps and curve risk, vanilla swaption pricing, local volatility, and selected dynamic models. The repository is a companion rather than a substitute for the literature, and it deliberately covers only part of the model universe reviewed here.

The central conclusion is simple but consequential: *there is no best interest-rate model independently of the payoff, calibration set, hedge instruments, market conventions, and numerical budget*. Modern desks do not select one equation to rule the franchise. They assemble a curve engine, volatility representation, dynamic model, and product-specific adjustment, then govern the interfaces between them. The remainder of the survey explains how that architecture emerged and how to reason about it.

2 | Foundations: Curves, Numeraires, Measures, and the Contracts a Model Must Price

2.1 | Discount Factors are the Primitive Language

Let $P(t, T)$ denote the time- t price of a default-free zero-coupon bond paying one unit at T . Under continuous compounding, the zero rate $R(t, T)$ and instantaneous forward rate $f(t, T)$ satisfy.

$$P(t, T) = e^{-(T-t)R(t, T)} = \exp \left[-\int_t^T f(t, u) du \right], \quad f(t, T) = -\frac{\partial \log P(t, T)}{\partial T}$$

The short rate is the diagonal value

$$r_t = f(t, t)$$

This identity already separates three tasks. A curve constructor estimates $P(0, T)$ from today's traded instruments. A dynamic term-structure model specifies how $P(t, T)$, $f(t, T)$, or another state evolves for $t > 0$. A derivative model combines that evolu-

tion with a payoff and a pricing measure. An exact fit of $P(0, T)$ says nothing by itself about future curve dynamics.

For an accrual period $[T_i, T_{i+1}]$ with year fraction δ_i , the single-curve simply compounded forward rate is

$$L_i(t) = L(t; T_i, T_{i+1}) = \frac{1}{\delta_i} \left(\frac{P(t, T_i)}{P(t, T_{i+1})} - 1 \right)$$

For a fixed-for-floating swap from T_α to T_β , define the annuity

$$A_{\alpha, \beta}(t) = \sum_{i=\alpha}^{\beta-1} \delta_i P(t, T_{i+1}),$$

and the single-curve par swap rate

$$S_{\alpha, \beta}(t) = \frac{P(t, T_\alpha) - P(t, T_\beta)}{A_{\alpha, \beta}(t)} = \sum_{i=\alpha}^{\beta-1} w_i(t) L_i(t), \quad w_i(t) = \frac{\delta_i P(t, T_{i+1})}{A_{\alpha, \beta}(t)}.$$

These relations explain why interest-rate modelling is intrinsically multi-dimensional. An equity option commonly refers to one spot. A swaption labelled "5Y $\times$ 10Y" is an option expiring in five years on a ten-year tenor swap, and therefore on a weighted collection of forward rates and discount factors. Changing the tenor changes the underlying swap rate. The swaption "surface" is therefore a cube indexed by expiry, underlying tenor, and strike, not a single-underlying surface $C(T, K)$.

2.2 | Bootstrapping Is Calibration, Not Clerical Preprocessing

Suppose instruments with market quotes $q_1, \dots, q_m$ have model prices.

$$V_j(P(0, T_1), \dots, P(0, T_m)).$$

A sequential bootstrap chooses curve nodes so that,

$$V_j(\mathbf{P}) = V_j^{\text{mkt}}(q_j), \quad j = 1, \dots, m/.$$

The system may reprice all pillars to machine precision, yet the curve between pillars remains an assumption. Linear interpolation in zero rates, log-linear interpolation in discount factors, cubic splines, and monotone-convex forwards can imply materially different forwards and hedge sensitivities. Hagan and West (2006) stress three desk-relevant criteria: *locality*, *smoothness*, and *economically acceptable forwards*. Locality means that perturbing one quote should not rewrite distant parts of the curve. Smoothness matters because discontinuous or oscillatory forwards contaminate pricing and risk. Shape preservation matters because a visually smooth zero curve can hide implausible forward spikes.

Parametric alternatives answer a different question. Nelson and Siegel (1987) write the zero yield for maturity $\tau = T - t$ as

$$y_t(\tau) = \beta_{0,t} + \beta_{1,t} \frac{1 - e^{-\lambda\tau}}{\lambda\tau} + \beta_{2,t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} e^{-\lambda\tau} \right)$$

The loadings approach 1, 0, and 0 at long maturities, while the second loading starts near one and decays, and the third is hump-shaped. This gives the familiar level, slope, and curvature interpretation. Svensson (1994) adds a second curvature component,

$$\beta_{3,t} \left(\frac{1 - e^{-\lambda_2 \tau}}{\lambda_2 \tau} e^{-\lambda_2 \tau} \right),$$

allowing two humps. Diebold and Li (2006) turn the factors into a forecasting system; Christensen et al. (2011) impose an arbitrage-free dynamic structure. Those uses are valuable for macro-finance, scenario design, relative value, and communication. They do not generally replace an exact-repricing curve used for trade valuation. Table 2 summarizes the practical differences among the main curve-construction and curve-representation methods.

In practice, the model developer should inspect not only repricing residuals but also one-basis-point quote bumps, forward curves, Jacobians from quotes to zero rates, and the stability of those Jacobians through time. The curve [modules](#)² in Feng (2026) provide a useful executable path from market instruments to discount factors, interpolation, forwards, present values, and DV01, while the Nelson–Siegel/Svensson [notebook](#)³ illustrates the separate role of parsimonious curve factors.

2.3 | Pricing by Numeraire

The fundamental pricing identity is best expressed using a numeraire. If $N_t > 0$ is a traded numeraire and $\mathbb{Q}^N$ its associated measure, then a payoff X_T has value.

$$V_t = N_t \mathbb{E}_t^{\mathbb{Q}^N} \left[\frac{X_T}{N_T} \right]$$

Under the money-market numeraire,

$$B_t = \exp \left(\int_0^t r_s ds \right),$$

discounted tradable prices are martingales under the risk-neutral measure $\mathbb{Q}$. Under the T -bond numeraire $P(t, T)$, ratios of tradables to $P(t, T)$ are martingales under the T -forward measure $\mathbb{Q}^T$. Under the swap annuity $A_{\alpha, \beta}(t)$, the associated forward swap rate is a martingale under the annuity measure $\mathbb{Q}^{A_{\alpha, \beta}}$.

This measure choice is the hidden engine behind market formulas. A caplet on $L_i(T_i)$, paid at T_{i+1} , is naturally priced under $\mathbb{Q}^{T_{i+1}}$, where L_i is a martingale. A European swaption is naturally priced under its annuity measure, where $S_{\alpha, \beta}$ is a martingale. The insight is due to the martingale approach to arbitrage pricing (Harrison and Pliska 1981) and is developed systematically in Björk (2004) and Musiela and Rutkowski (2005). The choice of numeraire does not change the economically correct price; it changes the drift and can turn a complicated expectation into a familiar option formula.

2.4 | Black and Bachelier are Market Languages

Under a lognormal assumption for a forward F_t under its natural measure,

$$dF_t = \sigma_B F_t dW_t,$$

the payer-option value with discounting scale D is,

$$V^{\text{Black}} = D [F_0 \Phi(d_1) - K \Phi(d_2)],$$

$$d_{1,2} = \frac{\ln(F_0/K) \pm \frac{1}{2} \sigma_B^2 T}{\sigma_B \sqrt{T}}$$

For a caplet $D = \delta_i P^d(0, T_{i+1})$; for a payer swaption $D = A_{\alpha, \beta}^d(0)$. Black (1976) therefore survives less as a full model of the economy than as an invertible map between European option prices and implied volatilities.

Under Bachelier's (1900) normal dynamics,

$$dF_t = \sigma_N dW_t$$

the payer-option value is,

$$V^N = D \left[(F_0 - K) \Phi(d) + \sigma_N \sqrt{T} \phi(d) \right], d = \frac{F_0 - K}{\sigma_N \sqrt{T}}$$

Normal volatility is measured in rate units per square root of time and remains well-defined when rates or strikes are negative. A shifted-lognormal convention instead replaces F and K by $F + s$ and $K + s$, retaining Black-like relative dynamics above a floor $-s$. Neither convention is “true.” They are coordinate systems for prices and risks. The important operational rule is that a volatility number is meaningless without its convention, shift, expiry, tenor, strike coordinate, day count, and annuity definition. The Black/Bachelier swaption [implementation](#)⁴ in Feng (2026) is an appropriate first laboratory because it makes the numeraire scale, forward-starting swap, implied-volatility inversion, negative-rate issue, and vega convention explicit before adding a full dynamic model.

3 | Why the Model Families Evolved: From Latent Short Rates to Observable Market Rates

3.1 | Equilibrium, No-Arbitrage, and Market Models

The most useful historical distinction is not one-factor versus two-factor. It is the transition from **equilibrium models**, to **no-arbitrage curve-fitting models**, to **market models**.

Equilibrium models begin with economically specified state variables under a physical or pricing measure and derive the term structure. Vasicek (1977), Dothan (1978), Brennan and Schwartz

TABLE 2 | Curve-construction and curve-representation methods.

Method	Fits liquid instruments exactly?	Local risk?	Smooth forwards?	Main strength	Main use	Main caution
Log-linear discount factors	usually	high	piecewise-discontinuous forward	robust, transparent	production bootstrap	kinks in instantaneous forwards
Linear zero rates	usually	high	limited	simple and familiar	basic curve engines	can distort long-dated forwards
Cubic spline	usually	often low	high visually	smoothness	government-bond fitting	overshoot and nonlocal DV01
Monotone-convex forwards	usually	comparatively high	high with shape control	desk-oriented compromise	OIS and projection curves	implementation details matter
Nelson–Siegel	no, unless augmented	n/a	very high	interpretation and parsimony	macro factors, forecasting	pricing residuals at pillars
Svensson	no, unless augmented	n/a	very high	additional curvature	central-bank and macro curves	parameter instability and local minima

(1979), and Cox et al. (1985) belong to this lineage. Their parameters have structural interpretations—mean reversion, long-run level, volatility—but the curve they generate need not equal today's observed curve. They are natural for economic analysis and useful as tractable building blocks.

No-arbitrage short-rate models reverse the priority. Ho and Lee (1986), Black et al. (1990), Hull and White (1990), and Black and Karasinski (1991) introduce time-dependent functions or calibrating trees so that today's term structure is reproduced. The initial curve is no longer a prediction; it is a boundary condition. This change made the models far more useful for relative pricing on sell-side desks.

HJM then takes the whole instantaneous forward curve as the state and derives its drift from volatility. Market models go one step further: they model the discrete forward LIBORs or swap rates that option markets quote. Their attraction is calibration alignment. A caplet is an option on a forward rate, and a swaption is an option on a swap rate, so why force those market quantities through an unobservable short rate if they can be modelled directly?

Figure 2 shows that later models are often specialisations or overlays rather than replacements. Hull–White is a Gaussian short-rate model and simultaneously a Markovian HJM specification. LMM is a discrete-tenor HJM market model. SABR can be attached separately to caplet or swaption marginals, or embedded into an LMM. Multi-curve and RFR frameworks change the curve objects and cash-flow definitions on which all these models operate.

3.2 | The Institutional Logic

Model evolution tracked the growth of derivatives markets. When liquid instruments were largely bonds and simple bond options, a one-dimensional short-rate tree was compelling. As cap and swaption markets developed, traders quoted implied volatilities on observable forwards and swap rates. Market models aligned state variables with these quotes. After volatility smiles became economically material, deterministic-volatility lognormal models could no longer describe cross-strike prices, motivating displaced diffusion, local volatility, SABR, and stochastic-volatility LMMs. After 2007, unsecured term rates, overnight collateral rates, and tenor basis ceased to be interchangeable, motivating multiple curves. After LIBOR cessation, the payoff rate itself became a compounded path of overnight fixings rather than a rate fixed at the start of the accrual period.

This account also explains persistence. Hull–White remains a workhorse not because its Gaussian short rate is a perfect empirical description, but because it fits the initial curve, supports fast trees/PDEs, generates complete future discount curves from low-dimensional states, and handles early exercise. SABR remains a quoting standard because four parameters communicate level, backbone, skew, and curvature and because an asymptotic formula makes calibration rapid. LMM remains important because it supplies joint forward dynamics, correlations, and path simulation. Model survival is therefore an engineering selection process:

THE ARCHITECTURE OF INTEREST-RATE MODELS

TERM-STRUCTURE PRICING PROBLEM

Different questions require different model layers

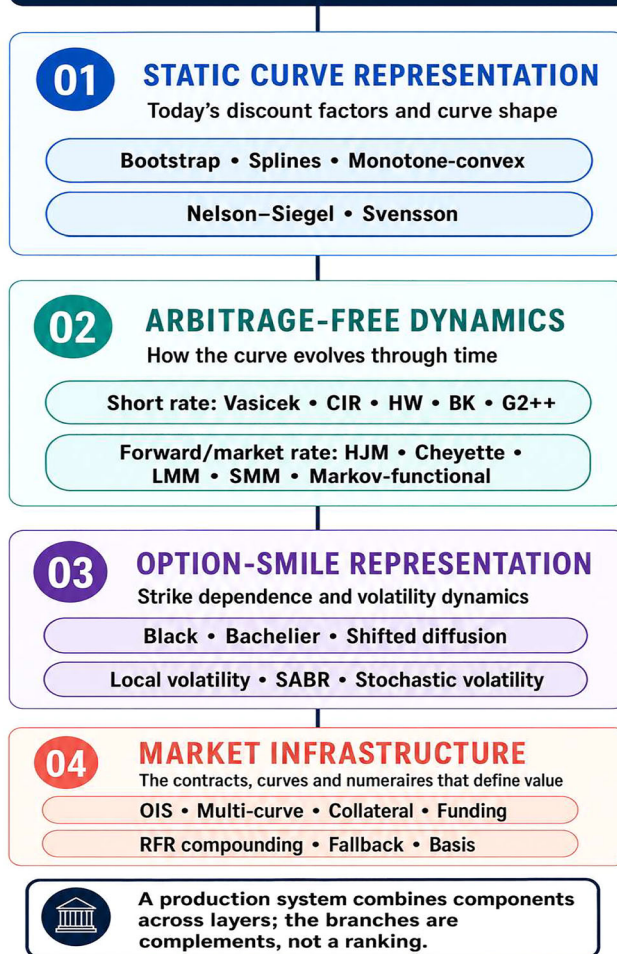


FIGURE 2 | The architecture and taxonomy of interest-rate models. The arrows indicate problem decomposition, not a ranking. A production system typically selects one or more components from every branch. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

the winning model is often the simplest one that preserves the market risks that dominate a particular book.

3.3 | The Scholar–Concept Lineage

Figure 3 condenses the main intellectual lineage. It is deliberately not a citation network based on counts. It shows how modelling ideas were transformed: Gaussian mean reversion was made curve-consistent; no-arbitrage forward-curve dynamics were made finite-dimensional; market-rate models were extended to smiles; and collateral/RFR changes rewrote the underlying curve system.

The taxonomy and lineage provide the foundation for detailed comparison. The next section examines the dynamic models; Section 5 adds smile; Section 6 maps both to products and

INTELLECTUAL LINEAGES IN INTEREST-RATE MODELLING

How foundational ideas became sell-side modelling architectures

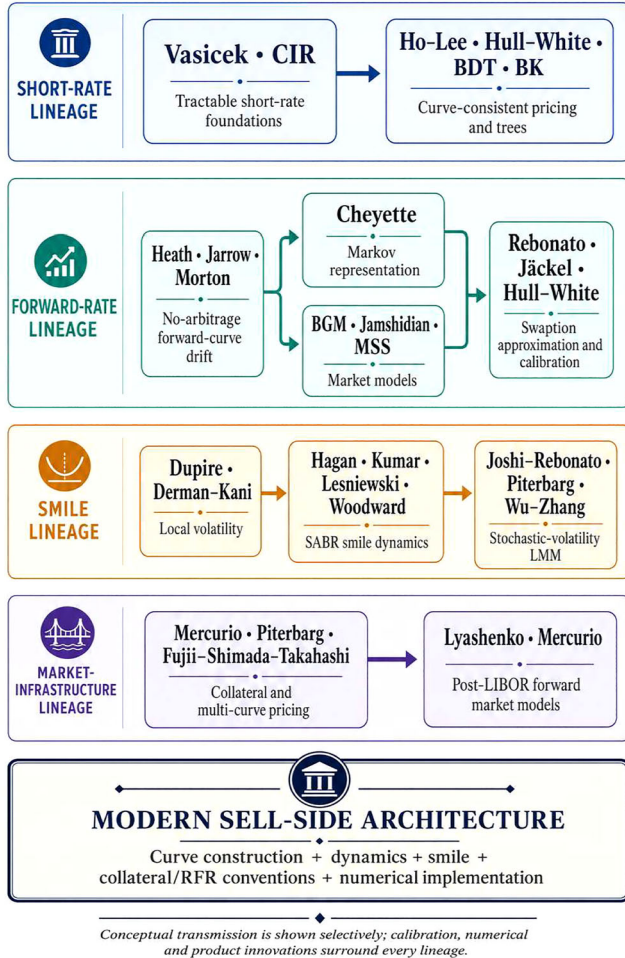


FIGURE 3 | Intellectual lineages in interest-rate modelling. The diagram highlights conceptual transmission among the main sell-side traditions. It is selective: many contributions refined calibration, numerics, and product applications around each node. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

numerical methods; Section 7 places them inside the modern collateral and RFR architecture.

4 | Dynamic Term-structure models

4.1 | One-Factor Equilibrium Short-Rate Models

The short-rate approach assumes that the entire term structure at time t is a function of one or a few Markov state variables, one of which is often r_t . Under the risk-neutral measure,

$$dr_t = \mu(t, r_t) dt + \sigma(t, r_t) dW_t$$

The value of a claim $V(t, r)$ with cash-flow-free interior then satisfies.

$$\frac{\partial V}{\partial t} + \mu(t, r) \frac{\partial V}{\partial r} + \frac{1}{2} \sigma^2(t, r) \frac{\partial^2 V}{\partial r^2} - rV = 0$$

This single equation reveals why short-rate models became foundational. Once μ and σ are specified and calibrated, bond prices, options, callable bonds, and early-exercise claims can be computed by analytic formula, lattice, or PDE in a low-dimensional state space.

Vasicek (1977) specifies the Ornstein–Uhlenbeck process.

$$dr_t = \kappa(\theta - r_t) dt + \sigma dW_t, \kappa > 0$$

Conditional rates are Gaussian:

$$r_T | r_t \sim \mathcal{N} \left(\theta + (r_t - \theta) e^{-\kappa(T-t)}, \frac{\sigma^2}{2\kappa} [1 - e^{-2\kappa(T-t)}] \right)$$

Zero-coupon bonds are exponentially affine,

$$P(t, T) = A(t, T) e^{-B(t, T)r_t}, \quad B(t, T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa}$$

The model's virtues are closed forms, mean reversion, and transparent factor duration. Its famous defect—negative rates—is not always a defect. Gaussianity was awkward when positivity was considered essential, but it became operationally convenient when several currencies experienced negative policy and swap rates. The deeper limitation is one-factor geometry: every curve move is generated by one state, so instantaneous yield changes are perfectly correlated across maturities.

Dothan (1978) uses a lognormal short rate,

$$dr_t = \sigma r_t dW_t,$$

which preserves positivity but lacks mean reversion and can generate implausible long-horizon dispersion. Brennan and Schwartz (1979) introduce two state variables, commonly a short and long rate, anticipating the need for slope dynamics. These models are historically important because they show the early trade-off: distributional realism and multiple factors reduce analytic and numerical convenience.

The Cox–Ingersoll–Ross model,

$$dr_t = \kappa(\theta - r_t) dt + \sigma \sqrt{r_t} dW_t,$$

links volatility to the level of rates (Cox et al. 1985). If

$$2\kappa\theta \geq \sigma^2,$$

the Feller condition makes zero unattainable. CIR retains affine bond prices and produces a noncentral chi-square transition law. Compared with Vasicek, it has economically appealing positivity and state-dependent volatility. Compared with later curve-fitting models, it has a serious desk limitation: a constant-parameter

CIR process cannot generally reproduce an arbitrary initial curve exactly. CIR++ remedies this by writing

$$r_t = x_t + \phi(t),$$

where x_t follows CIR and the deterministic shift $\phi(t)$ fits $P(0, T)$ (Brigo and Mercurio 2006). Positivity of x_t does not imply positivity of the shifted short rate.

Longstaff and Schwartz (1992) provide a two-factor general-equilibrium model in which one state captures the short rate and another its conditional variance. Such models matter for the economics of the term structure, but the sell-side lesson is broader: adding a factor is valuable only if it controls a risk that can be identified, calibrated, and hedged. A theoretically attractive latent factor that is weakly identified from liquid instruments can increase model risk rather than reduce it.

4.2 | Arbitrage-Free One-Factor Models: Fitting Today First

Ho and Lee (1986) specify, in continuous-time notation,

$$dr_t = \theta(t) dt + \sigma dW_t,$$

with a time-dependent drift chosen to fit the initial curve. The model is Gaussian and non-mean-reverting. Its importance lies less in current use than in the change of objective: exact term-structure fit became a design requirement.

Hull and White (1990) add mean reversion,

$$dr_t = [\theta(t) - ar_t] dt + \sigma dW_t$$

An implementation-friendly form is

$$r_t = x_t + \phi(t), \quad dx_t = -ax_t dt + \sigma dW_t$$

where the deterministic shift $\phi(t)$ fits the initial discount curve. With constant a and σ ,

$$\phi(t) = f(0, t) + \frac{\sigma^2}{2a^2}(1 - e^{-at})^2$$

under a common convention for the Gaussian factor. Bond prices remain exponential-affine:

$$P(t, T) = A(t, T) \exp[-B(t, T)r_t], \quad B(t, T) = \frac{1 - e^{-a(T-t)}}{a}$$

The parameter a controls how quickly shocks decay along the curve; σ controls the scale of uncertainty. A one-factor model cannot freely fit an entire swaption matrix with constant parameters. Desks therefore choose calibration baskets aligned with the product: co-terminal swaptions for a Bermudan, caplets for a callable floater, or a weighted combination. Time-dependent volatility improves fit but can create unstable piecewise parameters and awkward future dynamics. Exact calibration is not automatically superior to a smooth, economically interpretable approximation.

Hull–White’s key product advantage is conditional monotonicity. In a one-factor setting, a European option on a coupon bond can often be decomposed into options on zero-coupon bonds. Jamshidian (1989) finds a critical state r^* satisfying.

$$\sum_{i=1}^n c_i P(T, T_i; r^*) = K,$$

and uses the monotonicity of each $P(T, T_i; r)$ to write the coupon-bond option as a sum of zero-bond options with strikes $P(T, T_i; r^*)$. This “Jamshidian trick” gives fast European swaption pricing in one-factor models and underpins calibration.

Hull and White (1993) further develop bond-option pricing from bond-price evolution, reinforcing the model’s enduring analytic and implementation advantages.

The Black–Derman–Toy model is naturally expressed as a recombining lognormal short-rate tree calibrated to the yield and volatility term structures (Black et al. 1990). Its lattice representation made it influential for callable bonds and mortgage-style products. Black–Karasinski models the log short rate,

$$\text{dlog } r_t = [\theta(t) - \text{alog } r_t] dt + \sigma dW_t,$$

preserving positivity and mean reversion while fitting today’s curve through $\theta(t)$ (Black and Karasinski 1991). It is well suited to lattices but loses the Gaussian model’s simple bond-option formula. Positivity, once a major attraction, becomes a restriction when negative rates are possible; a deterministic shift can be added, again weakening literal positivity. Table 3 compares the principal short-rate models from the perspective of curve fit, tractability, and product use.

4.3 | Two-Factor Gaussian Models: Why G2++ Matters

One-factor models impose perfect instantaneous correlation across all zero-coupon yields. Empirical curve changes are dominated by a small number of factors, but not one. Litterman and Scheinkman (1991) identify the familiar level, steepness, and curvature directions. A two-factor model cannot reproduce every historical principal component, yet it can prevent a callable product from being priced as though the entire curve only shifts in parallel.

The G2++ model writes,

$$\begin{aligned} r_t &= \phi(t) + x_t + y_t \\ dx_t &= -ax_t dt + \sigma dW_t^{(1)}, \quad dy_t = -by_t dt + \eta dW_t^{(2)}, \\ d\langle W^{(1)}, W^{(2)} \rangle_t &= \rho dt \end{aligned}$$

The deterministic shift fits today’s curve, while different mean-reversion speeds create distinct maturity loadings. Bond prices remain exponential-affine in x_t and y_t , and European swaptions can be priced by low-dimensional integration (Brigo and Mercurio 2006). The cost is a more difficult calibration landscape: $(a, b, \sigma, \eta, \rho)$ can compensate for one another. Parameter stability and hedge behaviour may be more informative than the last decimal of calibration error.

TABLE 3 | Principal short-rate models from a sell-side perspective.

Model	Risk-neutral state equation	Curve fit	Distribution / factors	Main strengths	Main weaknesses	Natural products
Vasicek	$dr = \kappa(\theta - r)dt + \sigma dW$	no	Gaussian, 1F	closed form, transparent	negative rates; one-factor curve moves	pedagogy, benchmarks, simple bonds
CIR	$dr = \kappa(\theta - r)dt + \sigma\sqrt{r}dW$	no	square-root, 1F	positivity, affine pricing	poor exact curve fit; restricted smile	economic term structure, benchmarks
Ho-Lee	$dr = \theta(t)dt + \sigma dW$	yes	Gaussian, 1F	earliest exact-fit logic; simple tree	no mean reversion	historical benchmark
BDT	calibrated lognormal short-rate lattice	yes	lognormal tree, 1F	intuitive backward induction	calibration and grid dependence	callable bonds, embedded options
Hull-White	$dr = [\theta(t) - ar]dt + \sigma dW$	yes	Gaussian, 1F	fast, stable, analytic/tree/PDE friendly	limited smile and correlation geometry	Bermudans, callable bonds, XVA
Black-Karasinski	$d\log r = [\theta(t) - a\log r]dt + \sigma dW$	yes	lognormal, 1F	positivity and mean reversion	slower numerics; negative-rate difficulty	callable structures
CIR++	$r = x + \phi(t), x \text{ CIR}$	yes	shifted square-root, 1F	affine core plus exact curve	shift can defeat positivity	hybrid/counterparty models
G2++	$r = \phi + x + y, \text{ two Gaussian factors}$	yes	Gaussian, 2F	level/slope effects; analytic bond structure	calibration degeneracy; no intrinsic smile	Bermudans, callable and exposure

G2++ illustrates a recurring principle: **factor count is about conditional curve geometry, not merely in-sample fit**. For a European swaption, a one-factor model calibrated directly to that option can match its price. The second factor matters for products whose value depends on how several rates move between dates—for example, callable steepeners, Bermudans with changing underlying swaps, and exposure profiles of portfolios. Rebonato and Cooper (1995) show why apparently simple two-factor constructions can still impose unrealistic restrictions. Factor design should therefore be assessed through conditional correlation, exercise-boundary stability, and hedging instruments, not only through unconditional PCA.

4.4 | HJM: The Forward Curve Becomes the State

Heath et al. (1992) start from the instantaneous forward curve:

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) \cdot d\mathbf{W}_t$$

Under the money-market risk-neutral measure, absence of arbitrage imposes

$$\alpha(t, T) = \sigma(t, T) \cdot \int_t^T \sigma(t, u) du$$

This is the central insight of HJM. In most stochastic models, drift and volatility can be specified independently. HJM shows that for an arbitrage-free term-structure model, once the forward-rate volatility surface is chosen, the drift is uniquely determined by no-arbitrage. The modelling problem therefore becomes one of specifying and calibrating volatility rather than both drift and volatility. Since

$$P(t, T) = \exp \left[- \int_t^T f(t, u) du \right],$$

the model fits today's entire curve by construction: $f(0, T)$ is the initial condition.

HJM is a framework rather than one model. Deterministic volatility yields Gaussian forward dynamics and tractable bond prices. State-dependent or stochastic volatility adds realism but can destroy finite-dimensional Markov structure. The state is formally an evolving function $T \mapsto f(t, T)$, raising questions of existence, consistency, and numerical representation explored rigorously by Filipović (2001).

Cheyette (1996) shows how separable volatility structures can produce finite-dimensional Markov representations of HJM. A common quasi-Gaussian specification expresses bond-price dynamics through a small set of factors and accumulated variance states. This makes HJM compatible with PDE, lattice, or efficient Monte Carlo while retaining a direct connection to forward-curve volatility. The Cheyette class occupies an important middle ground: more curve-native than a short-rate model, more state-efficient than a generic HJM, and extensible with local or stochastic volatility.

The HJM viewpoint clarifies model relationships. Hull-White is not conceptually outside HJM; it corresponds to a particular separable Gaussian forward-volatility structure. Conversely, arbitrary HJM volatility need not admit a low-dimensional short-rate

representation. The practitioner's question is therefore not "short rate or HJM?" in the abstract, but whether the intended volatility kernel can be represented by a sufficiently small Markov state for the product and numerical method.

4.5 | The LMM/BGM Revolution: Model What the Market Quotes

Let tenor dates be $T_0 < \dots < T_N$, with forward rates,

$$L_i(t) = \frac{1}{\delta_i} \left(\frac{P(t, T_i)}{P(t, T_{i+1})} - 1 \right)$$

Under its own T_{i+1} -forward measure, a lognormal LMM specifies,

$$dL_i(t) = L_i(t) \sigma_i(t) \cdot d\mathbf{W}_t^{T_{i+1}}$$

Thus a caplet on L_i has an exact Black price under deterministic volatility. This direct consistency with the market's caplet convention was the central appeal of Brace et al. (1997), Jamshidian (1997), and Miltersen et al. (1997).

Joint simulation requires a common measure. Under the terminal measure $\mathbb{Q}^{T_N}$,

$$\frac{dL_i(t)}{L_i(t)} = - \sum_{j=i+1}^{N-1} \frac{\delta_j L_j(t)}{1 + \delta_j L_j(t)} \sigma_i(t) \cdot \sigma_j(t) dt + \sigma_i(t) \cdot d\mathbf{W}_t^{T_N}$$

The drift couples later forwards to earlier ones. Under a spot measure the drift has another form. This is not bookkeeping: a simulation can be locally plausible and globally arbitrage-inconsistent if the measure drift is implemented incorrectly.

Write,

$$\sigma_i(t) = \sigma_i(t) \mathbf{e}_i, \quad \mathbf{e}_i \cdot \mathbf{e}_j = \rho_{ij}$$

The model separates the caplet volatility term structure $\sigma_i(t)$ from the forward-rate correlation matrix ρ . A common parsimonious form is,

$$\rho_{ij} = \rho_\infty + (1 - \rho_\infty) \exp \left[-\beta |T_i - T_j| \right],$$

with a long-run floor ρ_∞ . Rebonato and Jäckel (2000) emphasise that a calibration correlation matrix must be positive semidefinite. Blindly fitting pairwise correlations can produce an invalid covariance matrix.

A swap rate is a weighted sum of forwards with stochastic weights. Freezing the weights and drifts gives the celebrated approximation for payer-swaption Black variance:

$$\sigma_S^2 T \approx \int_0^T \frac{\sum_{i,j=\alpha}^{\beta-1} w_i(0) w_j(0) L_i(0) L_j(0) \sigma_i(u) \cdot \sigma_j(u)}{S_{\alpha,\beta}^2(0)} du$$

Rebonato (1999a), Hull and White (2000), and Jäckel and Rebonato (2003) develop and test this link between caplet volatilities,

forward correlations, and swaption volatilities. It made calibration feasible: one could use cap prices to infer individual forward volatilities and swaptions to constrain correlations and combined volatility directions.

The approximation also exposes identification limits. Many combinations of forward volatilities and correlations produce nearly identical swaption prices. Rebonato (1999b) shows that simultaneous calibration is not simply a large nonlinear least-squares problem; it requires a parsimonious and stable parameterisation. A desk should ask which parameters are observable through available hedges. If a correlation mode changes an exotic price but barely affects any calibration instrument, it is a model-risk factor, not a well-estimated market parameter. Table 4 summarizes the main forward-curve and market-model families and their trade-offs for calibration and implementation.

4.6 | Markov-Functional and Swap-Market Ideas

If the main calibration instruments are swaptions but early exercise requires backward induction, a direct LMM creates a tension: market alignment comes with many states. Hunt et al. (2000) resolve part of the tension by expressing bond prices or rates as deterministic functions of a low-dimensional Markov process. The functional mapping is chosen to reproduce selected option prices. The driving process supplies tractable dynamics; the calibrated functions supply market consistency.

Markov-functional models are powerful for callable products because a one- or two-dimensional lattice can be used for backward induction while matching a swaption smile or set of marginals. The price is interpretive opacity. **The state variable may have no direct market meaning**, and smile dynamics follow from the calibrated mapping rather than from a transparent parameter such as SABR's ρ or ν . Extrapolation across expiries, tenors, and strikes requires care.

Swap market models attempt to make swap rates lognormal under their annuity measures, just as LMM makes forward rates lognormal under forward measures. Jamshidian (1997) sets out both LIBOR and swap market models and the associated measures. **The difficulty is joint consistency: overlapping swap rates share bonds, so assigning each an attractive marginal dynamic does not automatically create one arbitrage-free joint term structure.** In practice, LMM, Markov-functional, and low-dimensional HJM models offer different compromises among observable state variables, joint consistency, smile fit, and backward-induction speed.

4.7 | What the Dynamic Model Actually Identifies

The initial curve identifies no dynamic parameter. A strip of caplets largely identifies individual forward variances. A swaption matrix identifies weighted covariance combinations. It does not uniquely identify all pairwise forward correlations, future smile dynamics, or the conditional distribution of curve shapes after a large move. Exotic prices depend precisely on these weakly identified objects.

Rebonato (2002, 2004b) frames volatility and correlation as hedging objects rather than decorative statistical inputs. That viewpoint suggests a hierarchy:

calibrate liquid marginals $\rightarrow$ regularise joint dynamics
 $\rightarrow$ stress unspanned modes $\rightarrow$ validate hedge behaviour

The dynamic-model modules⁵ in Feng (2026) are most useful when read in this order. A Hull-White implementation teaches exact curve fit, mean reversion, path generation, and early-exercise-compatible numerics. An LMM implementation teaches measure-dependent drift, factor correlation, and multi-forward simulation. Combining them prematurely with stochastic volatility can conceal errors in discounting, schedules, or state evolution. A transparent implementation ladder is therefore not merely pedagogical; it is a model-risk control.

5 | Volatility Smiles: From Quotation Conventions to Stochastic Dynamics

5.1 | Why Deterministic Lognormal Volatility was not Enough

Black-76 turns each European option price into an implied volatility. If the lognormal model with deterministic volatility were a complete description, implied volatility would not depend on strike. In practice, caplet and swaption markets display skew and curvature that vary by expiry and tenor. The object to fit is

$$\sigma_{\text{imp}} = \sigma_{\text{imp}}(T_{\text{exp}}, T_{\text{tenor}}, K)$$

The cube is not a surface on one underlying. Each $(T_{\text{exp}}, T_{\text{tenor}})$ pair defines a different forward swap rate and annuity. This is why a direct transplant of equity local volatility is conceptually incomplete: fitting a local-volatility function for each swap rate does not automatically define arbitrage-consistent joint dynamics among overlapping swap rates.

Smile models perform at least three different jobs. They interpolate and extrapolate sparse strike quotes; they translate quoted normal or Black volatilities into prices and Greeks; and, for path-dependent products, they propose how the smile moves when rates and volatility move. The first job can be done by a static parametrisation. The third requires a dynamic model. Confusing the two is a major source of overconfidence.

5.2 | Displaced Diffusion and CEV-type Level Dependence

A displaced-diffusion forward follows

$$dF_t = \sigma(t)(F_t + s)dW_t,$$

so $F_t + s$ is lognormal and F_t is bounded below by $-s$. Equivalently, the dynamics may be viewed as a local-volatility specification with

$$\sigma_{\text{loc}}(F) = \sigma(t) \frac{F + s}{F} = \sigma(t) \left(1 + \frac{s}{F}\right)$$

TABLE 4 | Forward-curve and market models.

Family	State variable	Exact fit to today's curve	Vanilla calibration alignment	Dimensionality	Strength	Limitation
Generic HJM	$f(t, T)$	yes	indirect	potentially infinite	most general no-arbitrage forward framework	difficult Markovisation and numerics
Gaussian HJM / Cheyette	low-dimensional forward factors	yes	moderate	low	curve-native and numerically tractable	smile requires extension
LMM/BGM	discrete forwards L_i	yes on tenor grid	caplets exact; swaptions approximate	high before factor reduction	market-observable state and correlation control	drift cost, interpolation, smile and RFR issues
Swap market model	selected swap rates $S_{\alpha, \beta}$	locally	swaptions direct	high / consistency constrained	close to swaption quotes	joint no-arbitrage construction is difficult
Markov-functional	bonds or rates as functions of Markov driver	yes	can fit chosen vanillas strongly	low	calibration flexibility plus backward induction	model dynamics hidden in functional mapping
Generalised RFR FMM	forward-looking and backward-looking term rates	yes within framework	aligned with RFR products	moderate/high	models post-LIBOR payoff rates	more state and convention complexity

Hence the instantaneous volatility is no longer proportional to the forward level itself. When rates are low, the effective local volatility becomes larger, whereas Black dynamics correspond to a constant local volatility $\sigma_{\text{loc}}(F) = \sigma(t)$. The displacement therefore introduces a simple form of level dependence and can generate the skew commonly observed in interest-rate option markets.

Alternatively one may write:

$$dF_t = \sigma(t) [\beta F_t + (1 - \beta) F^*] dW_t,$$

with a displacement linked to F^* . This can be viewed as a local-volatility specification with

$$\sigma_{\text{loc}}(F) = \sigma(t) \frac{\beta F + (1 - \beta) F^*}{F} = \sigma(t) \left[\beta + (1 - \beta) \frac{F^*}{F} \right]$$

As F decreases, the second term becomes more important, so the effective volatility rises rates are low.

This alternative parameterisation makes the level dependence more explicit: the diffusion coefficient is no longer determined solely by the current forward level, but by a combination of the forward rate and a reference level F^* . As β approaches 1, the model approaches Black-style lognormal dynamics; smaller values of β weaken the dependence of volatility on the rate level.

Rubinstein (1983) introduced displaced diffusion as a flexible option model, and Andersen and Andreasen (2000) used nonlinear/CEV-style forward dynamics to extend LMM to skews.

Displacement is attractive because it is cheap, stable, and compatible with negative rates when the shift is sufficiently large. It changes the backbone—the relation between ATM volatility and the forward level—but deterministic displaced diffusion cannot independently control smile curvature or volatility-of-volatility. It is a useful baseline and often a component of richer stochastic-volatility LMMs (Joshi and Rebonato 2003).

5.3 | Local Volatility: Exact Static Fit, Fragile Dynamic Meaning

For a single tradable underlying with diffusion

$$dS_t = (r - q) S_t dt + \sigma_{\text{loc}}(t, S_t) S_t dW_t,$$

Dupire (1994) recovers local variance from a sufficiently smooth continuum of call prices:

$$\sigma_{\text{loc}}^2(T, K) = \frac{\partial_T C + (r - q) K \partial_K C + qC}{\frac{1}{2} K^2 \partial_{KK} C}$$

In an appropriate forward measure and forward-price coordinate, discounting and carry terms simplify, leaving the intuition

$$\sigma_{\text{loc}}^2 \propto \frac{\text{calendar} - \text{time change in option price}}{\text{risk} - \text{neutral density across strikes}}$$

Derman and Kani (1994) obtained the discrete implied-tree counterpart. The attraction is exact fit to a smooth arbitrage-free vanilla surface and a complete diffusion once the local-volatility function is specified.

The problems are amplified in rates. First, differentiation makes local volatility sensitive to noise and smoothing. Second, a swap-tion cube contains many related but distinct underlyings. Third, local volatility mechanically determines future smile dynamics, and those dynamics can be opposite to market behaviour. Hagan et al. (2002) made this critique central: **when the underlying changes, the local-volatility smile can move in the wrong direction, degrading hedges**. Fourth, independent local-volatility models for $1Y \times 5Y$ and $2Y \times 10Y$ swaptions do not define their joint dynamics.

Local volatility is therefore less dominant on rates desks than in single-underlying equity modelling. It remains useful as a diagnostic, a Markovian projection, a component of stochastic-local-volatility models, or a way to reproduce a particular marginal. Gatheral (2006) provides the broader volatility-surface treatment. Feng's (2026) local-volatility⁶ is best interpreted in precisely this limited but valuable way: it demonstrates price-surface construction, Dupire differentiation, interpolation, and Python/C++ consistency, while making clear why a collection of such marginals is not a complete swaption-cube model.

5.4 | SABR: The Market Language of Rate Smiles

The stochastic alpha-beta-rho model specifies, under the natural forward or annuity measure,

$$dF_t = \alpha_t F_t^\beta dW_t^{(1)}, \quad d\alpha_t = \nu \alpha_t dW_t^{(2)},$$

$$d \langle W^{(1)}, W^{(2)} \rangle = \rho dt$$

Its four parameters have unusually effective desk interpretations:

- α : current volatility scale and therefore largely the ATM level;
- β : backbone or elasticity of volatility to the forward level;
- ρ : rate-volatility correlation and a main driver of skew;
- ν : volatility-of-volatility and a main driver of smile curvature.

For $F \neq K$, Hagan et al. (2002) derive the widely used lognormal implied-volatility approximation

$$\sigma_B(F, K) \approx$$

$$\frac{\alpha}{\left((FK)^{\frac{1-\beta}{2}} \left[1 + \frac{(1-\beta)^2}{24} \log^2 \left(\frac{F}{K} \right) + \frac{(1-\beta)^4}{1920} \log^4 \left(\frac{F}{K} \right) \right] \right)^{\frac{z}{x(z)}}} [1 + \Gamma(F, K) T],$$

where

$$z = \frac{\nu}{\alpha} (FK)^{(1-\beta)/2} \log(F/K),$$

$$x(z) = \log \left(\frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho} \right),$$

and

$$\Gamma(F, K) = \frac{(1\beta)^2 \alpha^2}{24(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{(1-\beta)/2}} + \frac{(2 - 3\rho^2) \nu^2}{24}$$

At the money, the stable limiting expression is

$$\sigma_B(F, F) \approx \frac{\alpha}{F^{1-\beta}} \left[1 + \left(\frac{(1\beta)^2 \alpha^2}{24F^{2-2\beta}} + \frac{\rho\beta\nu\alpha}{4F^{1-\beta}} + \frac{(2 - 3\rho^2) \nu^2}{24} \right) T \right]$$

For negative-rate environments, shifted SABR replaces F and K by $F + s$ and $K + s$; normal SABR, closely related to $\beta = 0$, maps prices into Bachelier volatilities. The shift is not innocuous. It changes backbone, delta, and extrapolation. Two calibrations that fit the same quoted prices with different shifts can produce different smile Greeks.

The formula's speed, interpretability, and fit made SABR a market standard, but it is an asymptotic approximation rather than a globally exact law. Extreme strikes, long expiries, and parameter boundaries require safeguards. Oblój (2008) discusses corrections and asymptotic consistency; Antonov et al. (2013) develop more accurate treatments of the SABR distribution. Production calibration should enforce $\alpha > 0$, $\nu \geq 0$, $-1 < \rho < 1$, a controlled β , an admissible shift, and static-arbitrage checks on the resulting prices rather than merely on the parameters.

There is also a hedging subtlety. A naïve SABR delta differentiates the Black price while holding implied volatility fixed. A model-consistent delta includes the change in implied volatility with the forward. Bartlett (2006) shows that reparametrizing the volatility risk can yield a more robust “modified” delta. This distinction matters because SABR is used not only to interpolate prices but also to allocate vega, vanna, and volga across hedges.

5.5 | Why SABR Is Not a Full Term-structure Model

A separate SABR calibration for every expiry–tenor slice gives excellent marginal smiles. It does not, by itself, answer:

- how different forward and swap rates move jointly;
- how a future swaption smile conditional on a curve move will look;
- which short rate discounts a path-dependent cash flow;
- how an exercise decision interacts with smile and curve factors;
- how a CMS spread or callable exotic should be simulated consistently.

This is the most important conceptual boundary in practical rates modelling. SABR supplies a stochastic dynamic for one chosen forward or swap rate under one natural measure. A swaption

cube is a collection of such marginals on different underlyings. It becomes a full term-structure model only when embedded in a joint framework.

For a vanilla caplet or European swaption, a slice-by-slice SABR model may be exactly the right degree of complexity. For a smile-sensitive Bermudan or multi-rate exotic, one needs a stochastic-volatility short-rate/HJM model, an LMM with smile, a Markov-functional mapping calibrated to smiles, or a product-specific approximation.

5.6 | Stochastic-volatility Short-rate, HJM, and LMM Extensions

One route is to make the volatility of a Gaussian or quasi-Gaussian term-structure model stochastic. Schematically,

$$dx_t = -ax_t dt + \sqrt{v_t} dW_t^x$$

$$dv_t = \kappa_v (\theta_v - v_t) dt + \xi \sqrt{v_t} dW_t^v, d\langle W^x, W^v \rangle_t = \rho_{xv} dt$$

Such Hull–White/HJM stochastic-volatility models generate a complete future curve and smile dynamics, making them candidates for callable and hybrid products. Trolle and Schwartz (2009) construct a multifactor HJM model with stochastic volatility and quasi-analytical bond-option prices. The cost is a larger state space, harder calibration, and more expensive early-exercise numerics.

A second route extends LMM. Joshi and Rebonato (2003) combine displaced diffusion with stochastic volatility. Piterbarg (2003) introduces a forward-LIBOR stochastic-volatility model designed to fit a term structure of swaption smiles. Wu and Zhang (2006) develop an LMM with stochastic volatility and swaption approximations. These models seek three properties simultaneously:

caplet marginals + swaption covariance + smile/skew.

Achieving all three parsimoniously is difficult because each forward rate could in principle need its own volatility factor, skew, and correlation with other rates and volatilities. Factor sharing improves tractability but imposes restrictions across the cube.

The SABR/LMM approach developed by Rebonato et al. (2009) attaches SABR-style stochastic volatility to a forward-rate market model and focuses explicitly on pricing, calibration, and hedging of complex interest-rate derivatives. Rebonato et al. (2008) compare delta and vega hedging in SABR and LMM-SABR. Rebonato and Pogudin (2011) investigate whether caplet and swaption markets can be reconciled within a joint model. These contributions embody the sell-side modelling problem: matching each smile is easy relative to matching the cap and swaption markets, correlation structure, and hedge dynamics with one stable system. Table 5 places the main volatility and

TABLE 5 | Volatility and smile models.

Model / convention	Static vanilla fit	Dynamic content	Negative rates	Multi-rate consistency	Desk interpretation	Best use
Black-76	one ATM or strike at a time	deterministic lognormal marginal	no	none	high	quote and price vanillas
Bachelier	one ATM or strike at a time	deterministic normal marginal	yes	none	high	normal-vol markets and negative rates
Shifted lognormal	one quote at a time	lognormal above floor	yes, above $-s$	none	high	stable convention with chosen shift
Displaced diffusion / CEV-LMM	skew with few parameters	level-dependent diffusion	with displacement	good inside LMM	moderate/high	efficient skew-aware dynamics
Local volatility	exact smooth-surface fit	deterministic state-dependent vol	formulation-dependent	weak across swap rates	low/moderate	marginal fit, projection, diagnostics
SABR	excellent approximate slice fit	stochastic vol for one rate	shifted or normal version	none by itself	very high	caplet/swaption cube and smile Greeks
Stochastic-volatility HW/HJM	model-dependent	complete curve plus vol state	often yes	strong	moderate	smile-sensitive callable/hybrid products
SV-LMM / LMM-SABR	strong joint calibration target	forwards, correlation, stochastic vol	via shift/normal form	strong on tenor grid	high but complex	multi-rate smile-sensitive exotics

smile conventions alongside the dynamic extensions used for more complex interest-rate products.

5.7 | Static Fit, Future Smile, and the Unhedgeable Remainder

A useful decomposition of model error is

$$\begin{aligned} & \text{total model risk} \\ = & \text{static surface risk} + \text{dynamic smile risk} + \text{joint} - \text{rate risk} \\ & + \text{numerical risk} + \text{convention risk} \end{aligned}$$

Static surface risk concerns interpolation and extrapolation of today's quotes. Dynamic smile risk concerns how the surface moves. Joint-rate risk concerns correlation and conditional curve shapes. Numerical risk concerns discretisation, regression, and calibration. Convention risk concerns discounting, settlement, and benchmark definitions. An exact fit can eliminate only the first component at observed instruments; it may worsen the remaining components if the fitted parameters are unstable.

Rebonato (2003) asks which process can explain the observed dependence of swaption implied volatility on the underlying rate. That question remains a better model-selection test than minimum calibration error alone. A desk should compare sticky-strike, sticky-moneyness, shifted-lognormal, SABR, and model-implied smile moves against historical and hedging evidence, while recognising that historical dynamics under $\mathbb{P}$ and pricing dynamics under $\mathbb{Q}$ need not coincide. Model choice is ultimately a controlled extrapolation beyond traded vanilla prices.

6 | From Model to Price: Products, Calibration, Numerics, and Validation

6.1 | The Product Determines the Missing Distribution

The correct starting question is not “Which model is most sophisticated?” but “Which conditional distribution is missing after static replication and vanilla calibration?” A vanilla swap needs no stochastic rate model once curves and conventions are specified. A European swaption needs the terminal marginal distribution of one forward swap rate under its annuity measure. A Bermudan swaption needs conditional swap-rate distributions and an exercise policy at several dates. A callable range accrual needs the joint path of coupon fixings, discount factors, and exercise state. An XVA simulation needs portfolio-wide future values under legal netting, collateral, and possibly wrong-way risk.

The following hierarchy is useful:

linear cash flows	⇒ curves and conventions,
one European exercise	⇒ one marginal distribution,
several exercise dates	⇒ conditional dynamics,
several rates or path conditions	⇒ joint dynamics and correlation,
future portfolio value	⇒ global scenarios, revaluation and netting.

Figure 4 represents the complete pricing stack. The arrows go both ways in production. Market data flow downward into

THE COMPLETE INTEREST-RATE PRICING STACK

A model becomes operational only when every layer is coherent

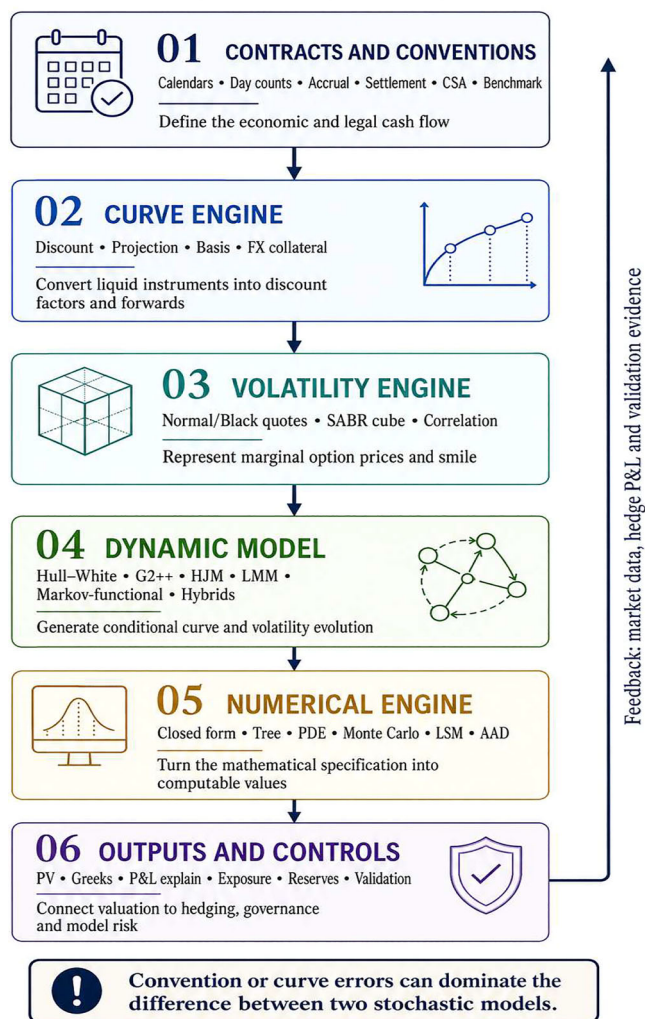


FIGURE 4 | The complete sell-side interest-rate pricing stack. *The model is only one component of the sell-side pricing stack. Convention and curve errors can dominate the difference between two stochastic models.* [Colour figure can be viewed at wileyonlinelibrary.com]

calibration and valuation; sensitivities, hedge P&L, and validation evidence flow upward and can force changes in quotes, regularisation, or model scope.

6.2 | Linear Products: The Curve Is the Model

For a collateralized fixed-for-floating swap with discount curve P^d and projection curve x , the floating leg value is no longer generally $P(t, T_\alpha) - P(t, T_\beta)$. It is

$$V_{\text{float}}(t) = N \sum_{i=\alpha}^{\beta-1} P^d(t, T_{i+1}) \delta_i F_i^x(t),$$

while the fixed leg is

$$V_{\text{fixed}}(t) = NK \sum_{i=\alpha}^{\beta-1} \delta_i P^d(t, T_{i+1})$$

Hence the multi-curve par rate is

$$S_{\alpha, \beta}^{x|d}(t) = \frac{\sum_{i=\alpha}^{\beta-1} \delta_i P^d(t, T_{i+1}) F_i^x(t)}{\sum_{i=\alpha}^{\beta-1} \delta_i P^d(t, T_{i+1})}$$

Nothing stochastic is required to price the deterministic cash-flow expectation implicit in a par-quoted swap once the forwarding and discounting curves are constructed consistently. The hard problems are quote selection, turn and meeting-date treatment, futures convexity, interpolation, basis consistency, and risk mapping. Calling this “model free” can be misleading because interpolation and instrument conventions are modelling assumptions. The swap and key-rate risk chapters⁷ in Feng (2026) demonstrate why a correct present value is not enough: fixed and floating legs, schedule generation, day counts, interpolation, curve bumps, steepeners, and flatteners must produce coherent component risk.

6.3 | European Options: Marginal Fit Usually Dominates

Caps and floors decompose into caplets and floorlets:

$$\text{Cap}(K) = \sum_i \delta_i P^d(0, T_{i+1}) \mathbb{E}^{T_{i+1}} \left[(L_i(T_i) - K)^+ \right]$$

Under deterministic-volatility lognormal LMM, each term has a Black price. Under normal dynamics it has a Bachelier price. With smile, a SABR calibration by caplet expiry or swaption expiry-tenor slice usually supplies the required marginal. The full path model is unnecessary if the payoff has one exercise date and depends on one rate.

This principle explains why a simple market formula can be more robust than a complete term-structure model for vanilla books. A full model introduces unneeded joint dynamics and numerical error. Its theoretical consistency is valuable only if the product uses it. Vanilla controls should therefore compare price-space residuals, implied-volatility residuals, vega-weighted residuals, and stability through time. A one-basis-point price error in a low-vega option is not economically equivalent to the same volatility error in a high-vega option.

6.4 | Bermudan Swaptions and Callable Bonds: early Exercise Changes Everything

At exercise date t_k , a Bermudan holder compares intrinsic value I_k with continuation value

$$C_k(\mathbf{X}_{t_k}) = \mathbb{E}^Q \left[e^{-\int_{t_k}^{t_{k+1}} r_s ds} V_{k+1} \mathbf{X}_{t_k} \right]$$

The decision is

$$V_k = \max \{I_k, C_k\}$$

Even if all co-terminal European swaptions are matched, the Bermudan price depends on their conditional joint dynamics and the shape of the exercise boundary. This is why Hull-White, G2++, Markov-functional, and LMM implementations can agree on calibration instruments but disagree on the exotic.

For one-factor Hull-White or Black-Karasinski, a recombining trinomial tree or one-dimensional PDE permits stable backward induction. G2++ adds curve-shape realism but requires two-dimensional integration, PDE, tree approximations, or Monte Carlo. An LMM aligns with market forwards and correlations but is high-dimensional. Andersen (2000) proposes an efficient exercise-boundary approach for Bermudans in multifactor LMM; Longstaff and Schwartz (2001) estimate continuation values by least-squares regression on basis functions. Broadie and Glasserman (1997) develop simulation estimators and bounds for American-style claims.

Least-squares Monte Carlo is conceptually simple but not self-validating. Basis functions, path count, exercise-state choice, in-sample bias, and extrapolation affect the boundary. A robust workflow reports lower-bound estimates from an admissible exercise policy, out-of-sample regression tests, convergence by paths and basis sets, and where practical an upper bound or benchmark. For callable bonds, notice periods, clean/dirty price conventions, accrued interest, call schedules, and credit spread treatment can matter more than moving from one Gaussian short-rate model to another.

The magnitude of this disagreement can be illustrated with a stylised calibration experiment. Consider a flat yield curve at 3% and flat 20% lognormal volatilities for the nine co-terminal European swaptions underlying a ten-year, annually exercisable payer Bermudan swaption struck at its forward par rate. We use the Hull-White model with piecewise-constant volatility and, for each fixed value of the mean-reversion parameter,

$$a \in \{-0.02, 0.01, 0.03, 0.05, 0.08, 0.12\}$$

bootstrap $\sigma(t)$ to the same nine European swaptions. The specification $a = -0.02$ is included as a stress case rather than as a conventional mean-reverting process. The result is in Table 6. Every model in the grid reprices all calibration instruments to within 0.1 volatility basis point, while the six models value the 1Y $\times$ 9Y European swaption to within 0.4 basis points of notional. Nevertheless, in this calibrated experiment, the Bermudan value increases monotonically from approximately 310 to 354 basis points of notional as a increases. The resulting 44-basis-point range represents approximately 14% of the lowest model value. The experiment therefore isolates what the European calibration basket does not identify: the intertemporal dependence governing continuation values and exercise decisions.

The mechanism is the identification gap of Section 4.7 made concrete. Mean reversion controls how fast rate shocks decay, hence the correlation between the swap rates observed at successive exercise dates. The co-terminal basket pins the value of exercising

TABLE 6 | Bermudan values across six vanilla-equivalent Hull–White calibrations.

Mean reversion a	Max. calibration error (bp vol)	Bermudan (bp)	Premium over European (bp)
−0.02	0.00	310.2	129.7
+0.01	0.00	319.5	138.9
+0.03	0.00	325.7	145.1
+0.05	0.06	332.0	151.3
+0.08	0.02	341.4	160.7
+0.12	0.00	354.1	173.2

Note: 10-year ATM payer Bermudan, callable annually from year 1, on a flat 3% curve with flat 20% lognormal co-terminal swaption volatilities; strike 3.0038% (the forward par rate). Each row is a Hull–White model with fixed mean reversion a and piecewise volatility bootstrapped to reprice the nine co-terminal Europeans to within 0.1 basis point of implied volatility. Values in basis points of notional; “premium” is the Bermudan value minus the same model’s 1Y × 9Y European. Deterministic integration engines; implemented in QuantLib (The QuantLib contributors 2026); the script (bermudan_mean_reversion.py) is provided as supplementary material.

at each date; the Bermudan is an optimal stopping problem against the joint law of rates across dates, and the joint law is exactly what the basket does not constrain. In this calibrated experiment, increasing a reduces the persistence of shocks across exercise dates and raises the Bermudan value. Desks manage this as a first-class exposure — the Bermudan book’s mean-reversion risk — and it is a concrete instance of the model-uncertainty interval of Cont (2006), whose governance role Section 6.11 takes up.

6.5 | Mortgage-Backed Securities: Early Exercise at Market Scale

The largest early-exercise problem in fixed income is not the Bermudan swaption book. It is the American prepayment option held by every fixed-rate mortgage borrower, securitised into agency mortgage-backed securities (MBS). The agency market had already exceeded \$5 trillion outstanding by the end of 2009, and Vickery and Wright (2013) document that the to-be-announced (TBA) forward market through which most agency MBS trade had an average daily trading volume second only to Treasuries among major US bond-market categories. The instrument extends the early-exercise problem of the previous subsection to market scale.

Exercise is behavioural rather than contractual. Let Π_t denote the pool’s surviving principal factor and SMM_t the single-month mortality—the fraction of principal prepaid in month t beyond the scheduled payment—quoted annually as a conditional prepayment rate:

$$CPR_t = 1 - (1 - SMM_t)^{12}, \quad \Pi_t = \Pi_{t-1} (1 - SMM_t) g_t,$$

where $g_t \in (0,1]$ is the scheduled-balance survival ratio: the fraction of principal that month t ’s scheduled payment alone would leave outstanding. A production prepayment model is an econometric hazard function, not an optimal-stopping rule. Schematically,

$$SMM_t = \psi(c - m_t) w(H_t) + u_t, \quad 0 \leq SMM_t \leq 1,$$

where c is the pool coupon, m_t the current mortgage rate, $\psi(\cdot)$ the S-curve mapping the refinancing incentive into prepayment speed, $w(\cdot)$ a burnout factor decreasing in the pool’s cumulative past incentive H_t , and u_t collects housing turnover, seasonality, and loan-age effects (Richard and Roll 1989; Schwartz and Torous 1989). In implementation, a bounded link function or equivalent constraint is required to ensure that SMM_t remains a valid probability. Stanton (1995) shows that this apparently irrational behaviour is consistent with optimal exercise under transaction costs and infrequent decision-making; production systems nevertheless estimate ψ and w statistically and re-estimate them as origination technology and policy change. In the language of this survey, an MBS pricing system couples a layer-2 dynamic model to a behavioural sub-model whose parameters are not determined by no-arbitrage restrictions. Burnout makes the hazard depend on the path of rates through H_t , not only on their current level—which is why Monte Carlo became the dominant flexible framework for path-dependent mortgage analytics, although lattice methods remain useful for simplified models and benchmarking.

The market’s spread language is built on top of this coupled system. The option-adjusted spread (OAS) of a pool trading at P_{mkt} is the constant ζ solving.

$$P_{\text{mkt}} = \mathbb{E}^Q \left[\sum_t CF_t \exp \left(- \int_0^t (r_s + \zeta) ds \right) \right],$$

with cash flows CF_t generated by the prepayment model along each simulated path. OAS is therefore the mortgage market’s analogue of implied volatility in its inversion role: it is a model-dependent number inferred from a market price and subsequently used as if it were a market observable, although its economic interpretation differs from that of implied volatility. Two desks running different prepayment models can report materially different OAS on the same pool at the same price, and Gabaix et al. (2007) show that OAS also embeds a priced premium for bearing prepayment risk, so even a perfectly specified prepayment model would not necessarily drive it to zero.

TABLE 7 | Product-to-model map.

Task / product	Minimum adequate model	Upgrade when needed	Dominant calibration set	Principal model risk
Vanilla IRS, OIS, basis swap	multi-curve bootstrap	stochastic model not required for PV	curve and basis instruments	interpolation, conventions, collateral
Cap/floor	Black/Bachelier + smile slice	SABR or CEV marginal	caplet strip	strike extrapolation and shift
European swaption	Black/Bachelier + SABR slice	full term-structure model only for consistency studies	expiry-tenor-strike cube	annuity/quote conventions, wings
Callable bond	HW/BK tree	G2++ or credit-rates hybrid	curve, bond options / swaptions	call behaviour, spread and factor risk
Bermudan swaption	HW/G2++/Markov-functional	LMM or stochastic-vol model	co-terminal swaptions and smile	conditional dynamics and exercise boundary
Agency MBS	multi-factor Monte Carlo engine + econometric prepayment model	risk-neutral (implied) prepayment frameworks	TBA prices, specified-pool pay-ups, swaption vols	prepayment specification and its rate sensitivity
CMS / CMS cap	SABR + static replication	multi-curve or dynamic model	swaption smile and CMS spread	wing extrapolation and convexity
Multi-rate path-dependent exotic	LMM / Cheyette	SV-LMM, LMM-SABR, SV-HJM	caps, swaptions, correlation proxies	unspanned joint and smile dynamics
XVA exposure	HW/G2++ or parsimonious HJM	multifactor/multi-currency joint engine	curves, vanillas, historical scenarios	future portfolio distribution and wrong-way risk

The hedging consequence is negative convexity at index scale. Because the S-curve steepens exactly where the refinancing option goes into the money, the pricing function flattens in rallies and steepens in sell-offs. Writing y for the level of the curve, effective duration is measured by finite difference under a parallel shift h :

$$D_{\text{eff}}(y;h) = -\frac{V(y+h) - V(y-h)}{2hV(y)}$$

It shortens when rates fall and extends when they rise, so effective convexity is negative near the current coupon—the coupon at which newly issued pools trade approximately at par. The aggregate duration of the MBS universe therefore moves with the level of rates, and the hedging flows of MBS holders and servicers can amplify rate moves—the feedback documented by Hanson (2014) and Malkhozov et al. (2016), visible in episodes such as the 2003 duration extension. The desk is structurally short an option it cannot buy back exactly; the practical hedge is a dynamically rebalanced position in swaps, swaptions, and Treasury futures whose effectiveness depends critically on one model quantity: the rate sensitivity of the fitted hazard,

$$\frac{\partial SMM_t}{\partial m_t}$$

Market infrastructure, in the sense of layer 4, is as decisive here as OIS discounting is for swaps. The TBA convention—forward sale of a pool for which the eligible collateral is specified through broad characteristics such as agency, coupon, and maturity, with the seller choosing the collateral ultimately delivered—creates the market's liquidity and embeds a delivery option analogous to the cheapest-to-deliver feature of a futures contract (Vickery and Wright 2013). Pools whose prepayment characteristics are valuable to the buyer trade over TBA at specified-pool pay-ups. The dollar roll—selling the front-month TBA and repurchasing it for settlement in the following month—is simultaneously the market's financing instrument and a prepayment-expectations trade, and it can trade “special” when collateral is scarce, much as repo does. A mortgage pricing system that ignores deliverability, the forty-eight-hour rule under which pool details are announced only two business days before settlement, and roll financing risks producing values for an abstraction rather than the instrument that is actually traded.

For the architecture of Table 1, MBS modelling is the stress test. It exercises every layer simultaneously and adds a sixth ingredient—the behavioural hazard—that the five-layer taxonomy of pure rates products does not need. That is the practitioner's justification for treating it here rather than scoping it out: the mortgage book is where the separation of layers is stretched furthest, and where the model governance of Section 6.11 spends a disproportionate share of its attention.

6.6 | CMS, Convexity, and Smile-Sensitive Payoffs

A constant-maturity-swap coupon pays a swap rate at a date that is not naturally associated with the swap annuity numeraire. The forward swap rate is a martingale under its annuity measure, but the CMS payoff is valued under a payment-date measure.

Therefore

$$\mathbb{E}^{T_P}[S_T] \neq S_0$$

in general. The difference is a convexity adjustment produced by the covariance between the swap rate and the annuity/discounting ratio.

Hagan (2003) develops the standard replication-based treatment of CMS swaps, caps, and floors. If $G(S)$ is the payoff adjusted for the change of numeraire, a twice-differentiable payoff can be represented schematically as

$$G(S) = G(K_0) + G'(K_0)(S - K_0) + \int_0^{K_0} G''(K)(K - S)^+ dK + \int_{K_0}^{\infty} G''(K)(S - K)^+ dK$$

The CMS value is thus linked to a continuum of payer and receiver swaptions across strikes. The far wings can matter materially, making SABR extrapolation and static-arbitrage control economic inputs rather than cosmetic curve choices. Mercurio and Pallavicini (2006) show how swaption skews and CMS adjustments can be treated consistently. In a multi-curve setting the discounting/forwarding split modifies the adjustment again.

For CMS spreads, range accruals, and callable CMS, a one-dimensional marginal is insufficient. The model needs the joint distribution of two swap rates, or the path of a rate relative to barriers, and possibly future exercise. Copulas or approximations can be adequate for isolated European spread options; LMM, stochastic-volatility LMM, or a carefully calibrated low-dimensional HJM is more natural for portfolios with shared curve and smile risks.

6.7 | Path-Dependent Exotics and Hybrid Books

Snowballs, target-redemption notes, callable range accruals, and Bermudan structures with multiple indices combine path dependence, early exercise, and several curve points. LMM was designed for this joint-forward problem, but the standard lognormal LMM lacks smile. Displaced diffusion, local-volatility LMM, stochastic-volatility LMM, and LMM-SABR trade computational cost for smile-sensitive dynamics.

A low-dimensional Hull-White or Cheyette model can be preferable when exercise and discounting dominate. A stochastic-volatility extension becomes relevant when smile materially changes future exercise or barrier probabilities. The practical choice depends on the hedge set. If the desk hedges primarily with co-terminal swaptions, the calibration should preserve those instruments. If a product has material mid-curve or spread risk, a model calibrated only to co-terminal marginals may conceal basis and correlation exposure.

Hybrid rates-FX, rates-inflation, or rates-credit products require additional joint factors and quanto drifts. The interest-rate component should remain no more complex than needed; otherwise the cross-asset correlation and calibration problem becomes unmanageable. A common production pattern is a Gaussian rates model for discount-factor tractability coupled to a

stochastic-volatility asset model. The modeller must then ensure measure changes and correlations are internally consistent.

6.8 | XVA: Dynamics for Exposure, Not Only Option Exercise

XVA engines simulate future portfolio value

$$V(t, \mathbf{X}_t)$$

over many dates and scenarios. Expected positive exposure and expected negative exposure are

$$\text{EPE}(t) = \mathbb{E}[V(t, \mathbf{X}_t)^+], \text{ENE}(t) = \mathbb{E}[-V(t, \mathbf{X}_t)^+]$$

The relevant model must generate complete future curves for all currencies and indices, support collateral and netting rules, and revalue large portfolios quickly. Hull–White and G2++ are common because low-dimensional Gaussian states make scenario generation and approximation efficient. LMM offers richer forward dynamics but can be expensive and awkward between tenor dates. A local Cheyette representation can bridge the two.

Exposure modelling is not identical to front-office exotic pricing. The objective is portfolio distribution and conditional revaluation rather than the exact price of one trade. Proxy functions, regressions, American Monte Carlo, or grids may be used. Piterbarg (2010) shows that collateral and funding change the valuation equations themselves, so exposure cannot be separated cleanly from discounting and legal terms. Wrong-way risk requires joint credit and market dynamics beyond the scope of a pure rates model.

6.9 | Numerical Methods are Part of the Model

A mathematical specification is incomplete until its numerical realization is defined. Glasserman (2004) provides the standard Monte Carlo foundations; Andersen and Piterbarg (2010) integrate analytics and numerics across interest-rate models. The principal choices are:

- **Closed form or approximation.** Fast and transparent; ideal for calibration and vanilla controls. The error is model/formula specific and can be systematic.
- **Tree or lattice.** Natural for one- or low-factor Markov models and early exercise. Grid geometry, state truncation, and calibration order matter.
- **PDE.** Accurate for low-dimensional diffusions, with strong boundary control. Cost grows rapidly with dimension.
- **Monte Carlo.** Flexible for many factors and path dependence. Early exercise, small Greeks, and nested expectations require special methods.
- **Fourier, quadrature, and asymptotics.** Valuable where transforms or low-dimensional integration exist; often used as benchmarks.

For an Euler simulation

$$X_{t+\Delta t} \approx X_t + \mu(t, X_t) \Delta t + \sigma(t, X_t) \sqrt{\Delta t} Z,$$

time-step error is part of model output. Exact transitions should be used where available, as in Gaussian Hull–White factors and CIR sampling. In LMM, frozen-drift and predictor–corrector schemes trade speed against drift accuracy. Rates must also be interpolated inside accrual periods; a tenor-grid model does not automatically specify the short rate or daily compounded overnight path.

Sensitivities are as important as values. Bump-and-revalue is transparent but costly and noisy. Pathwise and likelihood-ratio methods can improve Monte Carlo Greeks. Giles and Glasserman (2006) show that adjoint methods compute sensitivities to many curve or volatility inputs at a cost that is a small multiple of pricing, rather than one full revaluation per input. Adjoint Algorithmic Differentiation (AAD) does not remove modelling assumptions; it differentiates them efficiently. Discontinuities, calibration dependencies, and exercise policies still require careful treatment.

6.10 | Calibration Is an Inverse Problem

Let market prices be V_i^{mkt} , model prices $V_i(\theta)$, and weights w_i . A standard calibration solves

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_{i=1}^m w_i [V_i(\theta) - V_i^{\text{mkt}}]^2 + \lambda \mathcal{R}(\theta),$$

where $\mathcal{R}$ regularizes roughness, time variation, or deviation from prior parameters. Weighting by inverse bid–ask spread targets economically meaningful errors. Weighting in volatility space treats quotes symmetrically but can misrepresent low-vega instruments. Price-space and vega-weighted diagnostics should both be retained.

Calibration failures fall into three categories. **Under-identification** means many parameters produce the same vanilla prices. **Overfitting** means the model matches noise and generates unstable parameters or hedges. **Structural misspecification** means no parameter set can reconcile the cap, swaption, smile, and correlation information. Adding parameters addresses the first two only if the new directions are observable and regularised.

Figure 5 translates the above table into a decision sequence. “Need smile?” is not the first question for every trade: one first determines whether stochastic dynamics are needed at all.

6.11 | Validation Must Triangulate

A credible validation programme does not ask only whether the calibration optimiser converges. It triangulates:

1. **Analytic identities:** put–call parity, cap–floor parity, swap identities, numeraire consistency, and limiting cases.

A DECISION MAP FOR INTEREST-RATE MODELS

Select the least complex architecture that contains the payoff's material risks

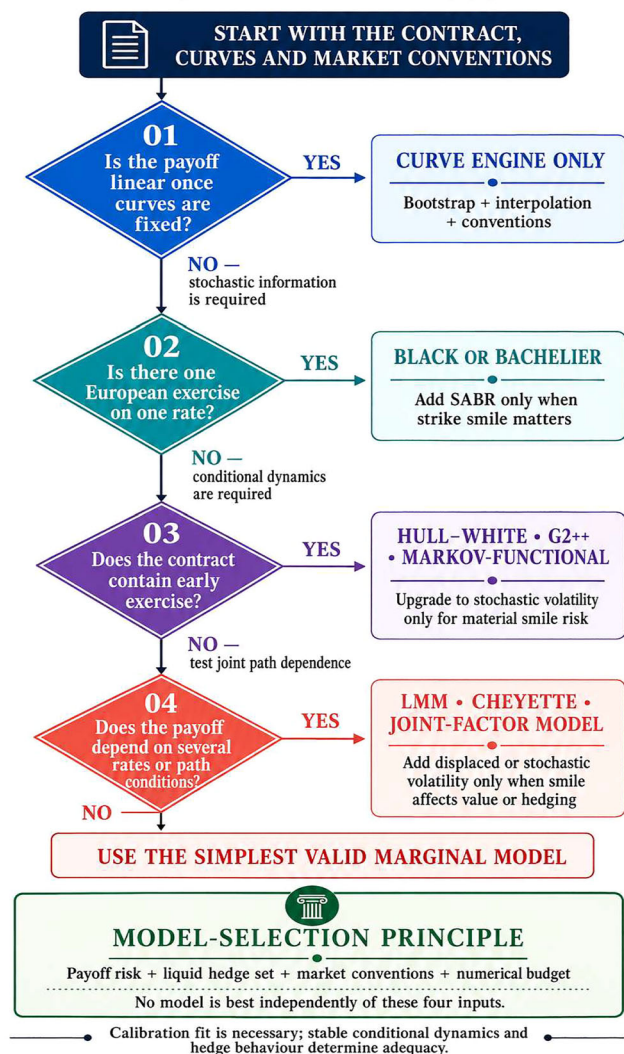


FIGURE 5 | A decision map for selecting interest-rate models. The recommended path seeks the least complex model that contains the payoff's economically material risks and can be calibrated to liquid hedges. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

- Independent numerics:** closed form versus tree, PDE versus Monte Carlo, or Python versus C++.
- Calibration reproduction:** prices, implied vols, bid-ask-scaled errors, parameter stability, and Jacobian conditioning.
- Risk reproduction:** finite differences versus analytic/AAD Greeks, curve-bucket aggregation, and vega mapping.
- Dynamic tests:** hedge P&L, smile moves, forward correlations, stress paths, and exercise-boundary behaviour.
- Implementation controls:** schedules, fixings, calendars, day counts, settlement lags, cash-flow signs, and collateral currency.

Table 8 makes the distinction between a calibration test and a model test explicit.

This triangulation is a Q-measure, desk-level notion of validation: it asks whether the model is internally consistent and hedges well against the instruments it was calibrated to. Section 8 turns to a complementary body of research that validates models against historical yield dynamics and against the joint pricing of interest-rate options.

This hierarchy also defines the pedagogical role of an open implementation. Feng's (2026) repository includes cross-language comparisons and explicit intermediate quantities—forward paths, discount factors, fixed and floating legs, curve shocks, implied volatilities—that make failures observable. A production library such as QuantLib is broader, but a compact educational codebase can be more effective for understanding because every model layer can be inspected. The best learning progression is to reproduce identities first, then prices, then Greeks, then calibration, and only then complex paths.

Validation is also an institutional arrangement, and the arrangement shapes which models exist. SR 11-7 (Board of Governors and OCC 2011) institutionalized model-risk governance in 2011; the revised interagency guidance SR 26-2 issued in 2026 subsequently replaced it with a more explicitly risk-based and proportionate framework (Board of Governors and OCC 2026). Under the revised framework, material models at large regulated banking organisations are typically recorded in an inventory, assigned accountable owners, and subjected to independent effective challenge. The negotiated output is rarely approval or rejection but approval *with limitations*: a product perimeter, monitoring thresholds, and a valuation reserve that converts residual model uncertainty into a balance-sheet number (Morini 2011). The cross-model valuation range in the framework of Cont (2006) provides an economically meaningful benchmark for, rather than a mechanical measure of, a model-risk reserve: in the Bermudan experiment reported in Table 6, vanilla-equivalent calibrations leave a 44-basis-point range in model value. Table 7 translates these calibration and model-risk considerations into a product-level map of minimum adequate models and possible upgrades. The reason model risk is largest exactly where calibration looks best is the unhedgeable-remainder lesson of Section 5.7: a model that fits every liquid quote has been given no chance to reveal what it extrapolates (Derman 1996). Governance cost thereby enters model selection on the same footing as the computational budget of Section 6.9, and the effective model set at any institution is the approved set, not the published set — one reason the diffusion of the Section 7 frameworks into production took years.

6.12 | Risk Ladders and P&L Explain: How the Desk Consumes a Model

The validation hierarchy above ends with hedge evidence; this subsection derives the two daily artefacts that generate it. For liquid instruments, the market sets the price; the model's daily work product is the risk report and the reconciliation of predicted against realised profit and loss.

A risk ladder is the portfolio's sensitivity bucketed by tradable market instrument: DV01—the present-value change for a one-basis-point move in a quoted rate—by curve tenor against par swaps; vega by expiry and underlying tenor across the swaption

TABLE 8 | Calibration and validation hierarchy.

Test level	Question	Example evidence	What passing does <i>not</i> establish
Unit / identity	Is the code internally correct?	parity, analytic limit, cash-flow reconciliation	realistic dynamics
Numerical convergence	Is the discretisation controlled?	time-step, grid, paths, seeds, basis functions	correct stochastic specification
Calibration	Does the model reproduce chosen liquid quotes?	residuals within bid-ask	unique parameters or exotic accuracy
Risk	Are local sensitivities stable and reconcilable?	bump vs analytic/AAD; P&L explain	tail and conditional dynamics
Dynamic / hedge	Does the model describe relevant moves?	backtested hedges and smile response	future regime robustness
Benchmark / challenger	Is model risk bounded by alternatives?	HW vs G2++ vs LMM; SABR vs normal/shifted	that either model is “true”

grid; and vega by strike where smile risk is tracked. The model, however, computes sensitivities to its own parameters, so the ladder represents a change of coordinates. If

$$\hat{\theta}(q)$$

is the calibration map of Section 6.10, sending the vector of market quotes q to the vector of model parameters $\hat{\theta}$, the market-quote risk of a position is

$$\frac{\partial V}{\partial q_j} = \sum_k \frac{\partial V}{\partial \theta_k} \frac{\partial \hat{\theta}_k}{\partial q_j}$$

In vector notation,

$$\nabla_q V = J_{\hat{\theta}}(q)^T \nabla_{\theta} V,$$

where $J_{\hat{\theta}}(q)$ is the calibration Jacobian. Every hedge ratio therefore inherits the calibration Jacobian—the same object whose conditioning appears in the validation list above.

Two consequences follow. First, curve construction is risk policy. A globally fitted spline can propagate a perturbation to a 30-year quote into the five-year risk bucket; the trader may then hedge artificial or nonlocal risk in an instrument they would not otherwise trade and incur an unnecessary bid-ask spread. This is the daily cost of the “exact repricing but unstable or nonlocal risk” failure mode of Table 1, and the reason production engines often prefer the local interpolation methods this survey already recommends (Hagan and West 2006).

Second, a hedge ratio is a joint statement of a model and a bumping convention. Bartlett’s modified SABR delta of Section 5.4 is the canonical example: bumping the forward rate F together with the backbone-consistent adjustment

$$d\alpha = \frac{\rho\nu}{F^\beta} dF$$

yields a systematically different delta from bumping F alone. The gap becomes a hedge error when the fitted backbone

does not represent the realised joint movement of the forward rate and volatility. Rebonato et al. (2008) extend the point to vega hedging in LMM-SABR. Two desks with identical calibrated models but different bump conventions—parallel versus principal-component curve shocks, or sticky-strike versus sticky-moneyness volatility shocks—can therefore run measurably different books against the same report format.

P&L explain closes the loop. Writing x_i for the curve buckets and σ_k for the volatility buckets, the desk decomposes each day’s realised P&L schematically as

$$\delta V = \Theta \delta t + \sum_i \Delta_i \delta x_i + \frac{1}{2} \sum_{i,j} \Gamma_{ij} \delta x_i \delta x_j + \sum_k \mathcal{V}_k \delta \sigma_k + NT + \varepsilon,$$

where Θ collects carry and time decay, Δ_i denotes first-order curve sensitivity, Γ_{ij} captures curve convexity and cross-bucket effects, $\mathcal{V}_k$ is the vega associated with volatility bucket k , NT records new trades and fees, and ε is the unexplained residual. The equation resembles a Taylor expansion; operationally, it functions as a daily ex post diagnostic of the model’s risk representation and hedge performance—the “dynamic” level of the hierarchy above, executed at production frequency.

A well-functioning book has a residual that is small, approximately unbiased, and economically interpretable. Its failures are often diagnostically useful. A residual related systematically to squared market moves, such as $(\delta x_i)^2$, can indicate missing or incorrectly bucketed convexity: an inaccurate Γ , omitted cross-gamma terms, or a ladder that is too coarse. A residual associated with volatility movements $\delta \sigma_k$ may indicate mis-specified smile dynamics, unstable calibration, insufficient vega bucketing, or an inappropriate sticky-strike or sticky-moneyness convention. A residual that jumps when a basis spread moves can reveal a risk factor that the model does not represent. In practitioners’ experience, this type of unexplained P&L was one indication of the limitations of single-curve pricing during 2007–2008, before the multi-curve framework surveyed in Section 7 became standard.

A small but persistently one-signed residual may be more dangerous than an isolated large error. It can signal a mis-marked parameter, an omitted carry component, or a slowly accumulating valuation adjustment that periodic validation might otherwise identify only after a considerable delay. Daily P&L explain can reveal such deterioration much earlier.

The desk-level summary for the model-selection doctrine of Section 9 is blunt. Among models that calibrate equally well—and the Section 6.4 table shows how little that constrains—selection should depend on the quality of the ladder and the explain: whether sensitivities land in tradable buckets, whether hedge ratios remain stable from day to day, and whether the unexplained residual is small, unbiased, and interpretable. A Greek that whipsaws generates transaction costs that marginal improvements in pricing accuracy may not repay. Exact vanilla calibration is a constraint; a clean explain is the objective.

7 | The Modern Market: Multi-Curve, Collateral, and the RFR Era

7.1 | The Single Curve Broke Because the Economics Changed

Before the global financial crisis, a common approximation used one curve both to project LIBOR cash flows and to discount them. Tenor basis spreads were small enough that the identities of Section 2 were operationally convenient. The crisis revealed material credit and liquidity differences across unsecured tenors, while collateralised derivatives were increasingly discounted at rates linked to collateral remuneration. The old equality

$$\text{discount curve} = \text{forwarding curve}$$

became economically false.

Mercurio (2009) develops new formulas and market models for the credit-crunch environment. Fujii et al. (2010) explain consistent construction of multiple swap curves with and without collateral. Bianchetti (2009) memorably summarizes the new practice as “two curves, one price.” Piterbarg (2010) derives valuation under collateral and funding from first principles. Ametrano and Bianchetti (2013) provide a broad practitioner treatment of multiple curves, while Henrard (2014) develops their foundations and implementation systematically.

Under perfect collateralisation in the payoff currency, a schematic valuation equation is

$$V_t = \mathbb{E}_t^{\mathbb{Q}^c} \left[e^{-\int_t^T c_s ds} X_T \right],$$

where c_t is the collateral remuneration rate and $\mathbb{Q}^c$ is associated with the collateralised discounting account under suitable assumptions. Incomplete collateral, asymmetric funding, thresholds, minimum transfer amounts, multiple eligible collateral currencies, and default break the simple linear representation. The resulting valuation can become institution-specific and nonlinear.

The multi-curve framework treats discount factors $P^d(t, T)$ and forwarding curves F^x for different indices/tenors as distinct but jointly constrained by OIS, IRS, and basis-swap quotes. Basis is not a cosmetic spread added after pricing. It changes par rates, option forwards, annuities, deltas, and model dynamics. Moreni and Pallavicini (2014) construct parsimonious HJM dynamics for multiple yield curves, illustrating that post-crisis consistency is a dynamic as well as a bootstrap problem.

7.2 | Multi-Curve Modelling Changes Drift and State

In a single-curve LMM, the forward rate is mechanically generated by a ratio of discount bonds. In a multi-curve system, a forwarding rate can be represented through a pseudo-discount curve or a multiplicative/additive spread relative to the discount curve. Its natural martingale measure and drift depend on the chosen tradable numeraire and model construction.

A useful multiplicative spread is

$$S^x(t; T_i, T_{i+1}) = \frac{1 + \delta_i F_i^x(t)}{1 + \delta_i F_i^d(t)},$$

where F_i^d is the simply compounded forward implied by the discount curve. Positivity of S^x is economically natural even when rates are negative. One can model discount-curve factors and basis-spread factors jointly, with correlations constrained by basis options or historical information. The number of unspanned modes grows quickly across tenors and currencies.

This is a central reason why “use LMM” is no longer a complete specification. One must state which forwards, which discount numeraire, which basis dynamics, which collateral currency, and which interpolation between tenor dates. A pre-crisis LMM applied mechanically to post-crisis cash flows can be mathematically elegant and economically wrong.

7.3 | Backward-Looking Overnight Rates

Risk-free-rate (RFR) benchmarks such as SOFR, SONIA, €STR, SARON, and TONA are overnight rates. A coupon over $[T_{i-1}, T_i]$ is often based on compounded daily fixings:

$$R_i^{\text{cmp}} = \frac{1}{\delta_i} \left[\prod_{k=1}^{m_i} (1 + \delta_{ik} r_{ik}^{\text{ON}}) - 1 \right]$$

Unlike an Interbank Offered Rate (IBOR) fixed near the beginning of the accrual period, most of R_i^{cmp} is unknown until near the payment date. Lockouts, lookbacks, observation shifts, payment delays, and caps/floors on daily or compounded rates make contract details integral to valuation.

Schrimpf and Sushko (2019) explain the economic transition from IBORs to overnight RFRs. Lyashenko and Mercurio (2019) define forward-looking and backward-looking term rates within one modelling framework; Lyashenko and Mercurio (2020) complete the generalized forward market model so that bond prices and

rates are jointly described. Their work shows that the LMM intuition survives, but the state must resolve the accrual interval and the setting-in-arrears nature of the payoff.

For a simple compounded RFR coupon, no convexity adjustment is needed merely because the rate is backward-looking if valuation uses the correct payoff and measure. Convexity appears when payment timing, averaging, futures daily margining, caps/floors, or nonlinear transformations change the relevant expectation. The modeller should derive the payoff from the contractual definition rather than retrofit an “RFR adjustment” to a LIBOR formula.

7.4 | What Survived LIBOR

LIBOR’s disappearance did not make Hull–White, HJM, LMM, or SABR obsolete. It changed their inputs and interpretation:

- Hull–White can model the OIS/RFR discount curve or a joint system with basis factors.
- HJM remains the general no-arbitrage language for forward curves, now potentially several curves.
- LMM’s discrete-forward insight extends to generalised forward market models for RFR-linked term rates.
- SABR remains a smile parametrisation for swaptions, caps, and RFR options, with normal or shifted conventions where appropriate.
- Black/Bachelier remain quotation mappings.

The transition demonstrates the paper’s main thesis. ***The market did not discover one superior stochastic differential equation. It changed the contracts and infrastructure, forcing the same model families to be re-embedded in a new architecture.***

7.5 | Production Architecture and the Role of Open Code

Table 9 maps the learning sequence to an implementation sequence and identifies where Feng’s (2026) repository can act as a companion. The mapping is intentionally narrower than the survey. The repository should not be presented as containing every model reviewed here; its value is that selected models are exposed from formula through code, numerical output, and cross-language comparison.

Open code has three scholarly benefits. It makes notation falsifiable: a sign error in an LMM drift or a date error in a swap schedule produces observable output. It makes numerical assumptions explicit: interpolation, root finding, discretisation, and random-number design cannot hide behind a formula. It also gives later researchers a stable pedagogical object to extend. These benefits justify repeated, task-specific references to the repository within the survey. They do not justify claiming production completeness. Clear scope is part of credibility.

8 | Empirical Validation and Diagnostic Testing: Testing Models Under the Physical Measure

While sell-side desks prioritize exact cross-sectional calibration under a pricing measure Q , model validation teams, buy-side risk managers, and central bank economists ask a different question: whether a term-structure model remains dynamically credible under the physical probability measure P , and whether it reproduces the joint empirical behaviour of yields and option-implied volatilities once embedded in a full pricing engine. This section surveys the diagnostic literature behind that question, organized around four validation dimensions (summarized in Table 10):

1. Time-Varying Risk Premia and Expectations Puzzles (P-Measure Dynamics)
2. The Failure of Volatility Spanning (Yields vs. Option Dynamics)
3. Joint Term-Structure and Option-Pricing Validation
4. Higher-Order Moments, Heavy Tails, and Residual Diagnostics

8.1 | Dimension 1: Time-Varying Risk Premia and Expectations Puzzles

A central benchmark for any dynamic term-structure model is whether its physical drift $\mu_P(t, x_t)$ can reconcile the empirical dynamics of bond yields, specifically the systematic failure of the Pure Expectations Hypothesis (PEH). Under P , the continuous-time evolution of latent state variables $X_t \in R^N$ is written as:

$$dX_t = \mu_P(X_t) dt + \Sigma(X_t) dW_t^P$$

Equivalent martingale pricing requires changing to the risk-neutral measure Q via Girsanov’s theorem, with Λ_t the vector of market prices of risk:

$$dW_t^Q = dW_t^P + \Lambda_t dt$$

$$\mu_Q(X_t) = \mu_P(X_t) - \Sigma(X_t) \Lambda_t$$

In the canonical Affine Term Structure Model (ATSM) framework of Duffie and Kan (1996) and Dai and Singleton (2000), zero-coupon yields are linear affine functions of the state variables, $Y_t(T) = A(T-t) + B(T-t)'X_t$. The Pure Expectations Hypothesis implies that in the Campbell and Shiller (1991) predictive yield regression:

$$Y_{t+\Delta}(T-\Delta) - Y_t(T) = \alpha_T + \beta_T \left[\frac{\Delta}{T-\Delta} (Y_t(T) - Y_t(\Delta)) \right] + \epsilon_{t+\Delta}$$

the coefficient should equal unity ($\beta_T = 1$) for all maturities. Instead, decades of empirical estimation (Campbell and Shiller

TABLE 9 | A staged curriculum for interest-rate modelling and implementation.

Stage	Theory to master	Implementation deliverable	Companion role of Feng (2026)	Beyond the current companion
1. Cash flows and curves	discounting, forwards, bootstrapping, interpolation	curve repricing and quote-to-zero Jacobian	bootstrap, interpolation, forward curve, DV01	multi-curve bootstraps across currencies and collateral
2. Curve factors	Nelson–Siegel–Svensson, PCA	factor fit and scenarios	parametric curve notebooks	affine macro-finance and arbitrage-free DNS
3. Linear rates products	schedules, IRS PV, par rate, curve risk	reconciled fixed/floating legs and shocks	IRS and swap-risk modules	futures, basis, cross-currency and inflation
4. Vanilla options	numeraires, Black/Bachelier, implied vol	caplet/swapion price and Greeks	forward-starting swaption and implied-vol inversion	full market data adapters and quote conventions
5. Smile	local volatility, SABR, arbitrage checks	calibrated smile slice/cube	local-vol and selected smile implementations	industrial cube cleaning, shifted/normal SABR variants
6. Low-factor dynamics	Vasicek/HW/G2++, trees/PDE	paths, bond options, Bermudan benchmark	selected Gaussian model components	production G2++, Markov-functional and stochastic vol
7. Market models	HJM, LMM drift, factors, correlation	multi-forward Monte Carlo	selected LMM components	multi-curve/RFR FMM, LMM-SABR, callable exotics
8. Portfolio engines	regression, AAD, exposure, XVA	Greeks and future-value profiles	building blocks and validation examples	collateral/netting engines and distributed computation

TABLE 10 | Empirical validation matrix: testing term-structure models under P and Q.

Validation dimension	Primary empirical issue / Failure mode	Key diagnostic landmark studies	Architectural and operational remedy
1. Time-Varying Risk Premia	Campbell–Shiller expectations puzzle ($\beta_T < 0$); rigid coupling of factor variance and risk premia under P .	Dai and Singleton (2001); Duffee (2002); Cochrane and Piazzesi (2005)	Adopt Essentially Affine specifications (AO(N) with unconstrained Λ_t) for multi-horizon scenario generation and yield forecasting.
2. Volatility Spanning	Yield factors fail to span swaption/cap implied volatilities; ATSMs misfit volatility dynamics under P .	Collin-Dufresne and Goldstein (2002); Bikbov and Chernov (2009); Creal & Wu (2015)	Decouple Layer 2 (Yield Dynamics) from Layer 3 (Smile Layer); overlay SABR, local volatility, or Markov-functional models onto the yield engine.
3. Joint Term-Structure / Option Validation	Yield-calibrated models misprice caps/swaptions; PCA shows yield factors explain <60% of implied-vol variance.	Jagannathan et al. (2003); Longstaff et al. (2001); Heidari and Wu (2003); Li and Zhao (2006)	Confirms the five-layer architecture: maintain separate, jointly-calibrated curve, dynamics, and smile layers rather than one spanned process.
4. Moments and Tail Risk	Continuous ATSMs understate tail risk and produce serially correlated calibration residuals.	Feldhütter (2016); Hamilton and Wu (2012)	Incorporate jump-diffusions or regime-switching for XVA/tail-risk engines; monitor residual autocorrelation (ρ_1) for model risk.

1991; Dai and Singleton 2001) find β_T significantly below 1, becoming increasingly negative as maturity lengthens—the Campbell–Shiller expectations puzzle. Dai and Singleton (2001) show why: standard “Completely Affine” models parameterize the market price of risk as $\Lambda_t = S_t \lambda$ (below), which tightly links factor variance to risk premia, forcing β_T to remain positive and making it impossible to fit cross-sectional yield volatility under Q and time-varying risk premia under P simultaneously:

$$\Lambda_t = S_t \lambda$$

Duffee (2002) resolves this with “Essentially Affine” models, adding a variance-independent term to the price of risk:

$$\Lambda_t = S_t \lambda^0 + S_t^- \Lambda^1 X_t$$

This lets Gaussian ATSMs match the negative Campbell–Shiller slope without the out-of-sample forecasting breakdown Duffee termed “engine failure.” Cochrane and Piazzesi (2005) sharpen the benchmark further: a single tent-shaped combination of forward rates (the CP factor, $F_t = \gamma' f_t$) predicts annual excess bond returns across all maturities with $R^2 > 0.35\text{--}0.40$. An N-factor model validated for scenario generation should reproduce this single-factor concentration of return predictability.

8.2 | Dimension 2: The Failure of Volatility Spanning (Yields vs. Option Dynamics)

Standard N-factor short-rate models (Vasicek, CIR, Hull–White, G2++) and low-dimensional ATSMs assume volatility spanning: the state variables X_t that price zero-coupon bonds also determine the conditional variance of yield changes,

$$\text{Var}_t(dY_t) = B(T - t)' [\Sigma(X_t) \Sigma(X_t)'] B(T - t)$$

implying option-implied volatilities should be redundant with the yield curve itself. Collin-Dufresne and Goldstein (2002) and Bikbov and Chernov (2009) reject this: yield-curve principal components (Level, Slope, Curvature) explain less than 20% of the variation in interest-rate option implied volatilities. Most rate volatility dynamics are Unspanned Stochastic Volatility (USV)—factors that leave bond prices nearly unaffected but strongly drive option prices and tail risk. Creal and Wu (2017) formalize the resulting trade-off: models estimated on yields alone fit the zero curve well but misfit realized/option volatility, while USV models that restrict volatility factors out of the bond-pricing loadings capture P-measure volatility dynamics at the cost of Q-measure yield-fitting error.

This literature is the direct empirical justification for decoupling Layer 2 (Arbitrage-Free Yield Dynamics) from Layer 3 (Volatility-Smile Representation) in the five-layer architecture of this survey: because volatility is empirically unspanned, sell-side desks overlay dedicated smile models (SABR, local volatility, Markov-functional) onto the yield engine rather than forcing one spanned process to fit both.

8.3 | Dimension 3: Joint Term-Structure and Option-Pricing Validation

A closely related empirical literature asks a sharper version of the spanning question: when a term-structure model is extended into a full derivatives-pricing engine, can it match both the cross-section of zero-coupon yields and the empirical moments of the cap and swaption volatility surface simultaneously? The answer, across three complementary strands of evidence, is consistently NO for low-dimensional spanned models.

Joint fitting studies find that yield-calibrated models misprice options even when the yield fit is excellent. Jagannathan et al. (2003) show that a three-factor CIR model fits the LIBOR/swap curve well but generates option pricing errors that are large relative to the bid-ask spread once tested against caps and swaptions. Longstaff et al. (2001) find that a string market model requires four to five factors to match the correlation structure priced into the swaption cube, and that low-dimensional yield models systematically underprice long-tenor swaptions. Driessen et al. (2003) show that adding factors improves cross-sectional option fit but can degrade out-of-sample hedge stability—an early empirical warning about the parameter drift that also motivates the diagnostics in Section 6.11.

Volatility-moment studies quantify the shortfall directly. Heidari and Wu (2003) apply principal component analysis jointly to the yield curve and the swaption implied-volatility matrix: yield components explain over 99% of yield variance but less than 60% of implied-volatility variance, requiring three additional, largely yield-independent factors. Trolle and Schwartz (2009) show that an HJM model with unspanned stochastic volatility matches conditional volatility, skew, and out-of-sample option pricing errors far better than affine or deterministic-volatility alternatives—consistent with Bikbov and Chernov (2009), who find that forcing volatility to be spanned by yields produces large, serially correlated option pricing errors.

Hedging-based studies confirm the same conclusion from a different angle. Li and Zhao (2006) show that models with unspanned stochastic volatility achieve substantially lower out-of-sample hedging errors on caps than spanned alternatives, and Andersen and Andreasen (2000) show that standard lognormal LMM cannot match the empirical strike-dependence of caplet and swaption prices without an explicit local- or stochastic-volatility layer.

Taken together, this literature is a direct empirical vindication of the paper's central architectural claim: no single spanned dynamic equation reproduces both the yield curve and the option surface. That is precisely why front-office desks maintain curve construction, arbitrage-free yield dynamics, and volatility-smile representation as separate, jointly-calibrated layers rather than one monolithic process.

8.4 | Dimension 4: Higher-Order Moments, Heavy Tails, and Residual Diagnostics

Risk management and stress-testing applications (VaR, Expected Shortfall, XVA simulation) depend on higher-order moments—

skewness and kurtosis—that first- and second-moment calibration does not discipline. Feldhütter (2016) compares sample moments of historical yield changes against moments simulated from calibrated $A_m(N)$ affine models: Gaussian models, $A_0(N)$, generate zero skewness and zero excess kurtosis by construction, understating heavy tails, while volatility-dependent models ($A_1(N)$, $A_2(N)$) still fail to match observed tails without producing unrealistic yield curves. This motivates jump-diffusion (Johannes 2004) or regime-switching (Dai et al. 2007) extensions for long-horizon XVA and tail-risk applications.

A related diagnostic concerns the residuals of the calibration itself. Model calibration minimizes a cross-sectional loss between market and model-implied prices,

$$\min_{\theta} \sum_{j=1}^m w_j \left| V_j^{mkt} - V_j(\theta) \right|^2$$

and standard practice treats the resulting errors as i.i.d. Gaussian noise. Hamilton and Wu (2012) show that calibration residuals in practice exhibit severe serial correlation ($\rho_{-1} > 0.90$), which signals that the model's state space is structurally misspecified rather than merely noisy—complementing the calibration-reproduction check of Table 8 with an explicit statistical test for when a stable-looking calibration is nonetheless missing a state variable.

9 | Synthesis: A Model-Selection Doctrine and Research Agenda

9.1 | Seven Questions Before Choosing a Model

The history can be compressed into seven questions:

1. **What is the contract?** Identify the benchmark, accrual, settlement, collateral, optionality, and legal cash-flow definition.
2. **Which curves are required?** Separate discount, projection, basis, FX-collateral, and credit components.
3. **What remains random After static replication?** One terminal rate, several rates, an exercise policy, or a full portfolio path?
4. **Which market instruments span the risk?** Match calibration to actual hedge instruments, not to every available quote indiscriminately.
5. **Does smile matter to value or only to reporting?** Add SABR or stochastic volatility when strike dependence changes price or hedge materially.
6. **Which numerical method respects the state dimension and payoff?** Closed form, lattice, PDE, or Monte Carlo is part of the model choice.
7. **Which outputs are weakly identified?** Correlation, future smile, basis dynamics, and tails require stress tests and reserves.

These questions produce a disciplined principle:

Choose the least complex model that represents all material, hedgeable payoff risks under the correct market conventions.

“Least complex” does not mean simplistic. A multi-curve LMM-SABR can be the least adequate model for a smile-sensitive callable range accrual. A deterministic curve is the least adequate—and best—model for a vanilla swap PV. Complexity is relative to the missing distribution.

9.2 | Persistent Misconceptions

Five misconceptions deserve explicit rejection.

First, **exact curve fit does not imply a good curve model**. Interpolation can create unstable forwards and hedge risk even when every pillar reprices.

Second, **exact vanilla calibration does not identify exotic dynamics**. Marginal prices constrain only projections of the joint process.

Third, **positive rates are not an unconditional virtue**. CIR and lognormal models encode a boundary; Gaussian and shifted models encode another. The relevant question is whether the boundary is plausible for the currency, horizon, and stress regime.

Fourth, **SABR is not a yield-curve model**. It is an exceptionally useful marginal stochastic-volatility model and quotation language. Joint use requires an embedding.

Fifth, **a modern interest-rate model is not separable from infrastructure**. Collateral, basis, RFR compounding, and settlement determine the tradables and numeraires to which no-arbitrage arguments apply.

9.3 | Where Research Remains Valuable

The most useful future work is unlikely to be another unconstrained high-dimensional diffusion that fits the same cube slightly better. Four areas are more consequential.

The first is **identification-aware calibration**: regularisation and uncertainty quantification should distinguish parameters identified by vanilla instruments from extrapolative assumptions. The second is **dynamic validation**: hedge tests should evaluate conditional curve and smile moves, not only end-of-day fit. The third is **cross-layer consistency**: curve interpolation, smile extrapolation, model dynamics, and collateral conventions should be validated as one pipeline. The fourth is **transparent computational acceleration**: adjoints, algorithmic differentiation, surrogate models, and machine learning can reduce cost, but they should preserve arbitrage identities, sensitivities, and auditability.

Rebonato (2023) argues persuasively that the development of risk-neutral forward-rate dynamics can be understood only in

its trading context. Extending that insight, the durable unit of innovation is the pricing architecture. A model succeeds when it converts the market’s liquid information into stable values and hedges for non-liquid claims at an acceptable computational and governance cost.

9.4 | Conclusion

Interest-rate modelling evolved through a sequence of changing questions. Vasicek and CIR made the short rate tractable. Ho–Lee, BDT, Hull–White, and Black–Karasinski made today’s curve a calibration target. HJM made forward-curve drift a no-arbitrage consequence of volatility. LMM aligned dynamics with traded caplet and swaption conventions. Local volatility and SABR addressed strike-dependent prices; stochastic-volatility HJM and LMM models sought joint smile dynamics. Multi-curve, collateral, and RFR frameworks then changed the objects being modelled.

No member of this sequence eliminated its predecessors. Each remains useful where its state variables, calibration instruments, and numerics match the payoff. The correct mental model is therefore modular:

curve engine + volatility surface + dynamic model +
market conventions + numerical method.

For students, this architecture gives a learning path. For researchers, it locates genuine gaps in identification, dynamics, and implementation. For practitioners, it provides a model-selection and validation discipline. Theory becomes durable when it is connected to contracts, quotes, computation, and hedge evidence. That connection—rather than the search for a universally strongest equation—is the organising logic of modern interest-rate modelling.

Author Contribution:

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Conflicts of Interest

No funding was received for this research. The author declares no conflicts of interest. The discussion of the open companion repository is pedagogical. No commercial product or service is endorsed.

Data Availability Statement

Selected implementations discussed in the survey are publicly available in Xuan Feng’s *Rates-Pricing-Quant-Library*: <https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library>.

Generative AI Statement

Generative artificial intelligence tools were used solely to refine the language and improve the clarity of selected sentences, and to produce digital versions of the figures from detailed structures and layouts designed by the author. The conceptual framework, arguments, literature selection, interpretation, figure content, and final verification were undertaken by the author. All AI-assisted text and visual outputs were critically reviewed, revised, and approved by the author, who takes full responsibility for the content of the manuscript.

Endnotes

¹ Available at: <https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library>

² Available at: https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library/tree/main/01_yield_curve_bootstrap

³ Available at: https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library/tree/main/02_parametric_yield_curve_models

⁴ Available at: https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library/tree/main/06_black_swaption_pricing

⁵ See Chapter 09, 10, 13, and 14 from <https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library>

⁶ Available at: https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library/tree/main/07_local_volatility_and_dupire

⁷ See Chapter 04-06 from <https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library>

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Appendix A

Extended Road Map of the Survey

Because this survey is deliberately comprehensive and has a partly textbook-like character, it is longer and more structurally layered than a conventional literature review. The following extended table of contents is therefore provided as a navigational aid. It enables readers—especially doctoral students, early-career quantitative analysts, practitioners, and model validators—to move directly between the historical evolution of interest-rate modelling, its mathematical foundations, major model families, product applications, calibration and validation practices, and the accompanying implementation material. Readers may consequently approach the article either sequentially, as a systematic introduction to the field, or selectively, as a reference work for a particular model, product, or practical modelling problem.

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 - 2.2 Bootstrapping is calibration, not clerical preprocessing
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- 5.5 Why SABR is not a full term-structure model
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 - 9.3 Where research remains valuable
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Appendix B

GitHub Rates-Pricing-Quant-Library: A Reproducible Learning Architecture for Interest-Rate Modelling

The open-source [Rates-Pricing-Quant-Library](#) (advertised in Figure A0) is designed as a computational companion to this survey. It translates the historical and conceptual development of interest-rate modelling into a fifteen-chapter implementation pathway, beginning with curve construction and forward-rate instruments, advancing through vanilla option pricing, volatility smiles and term-structure dynamics, and concluding with Bermudan exercise, multi-forward simulation, stochastic-volatility hybrids, and counterparty exposure/XVA. The repository emphasises transparent formulas, front-office interpretation, reproducible numerical experiments, and cross-language validation rather than black-box implementation.

All chapters combine a narrative Jupyter notebook with reusable Python modules, selected C++ implementations, sample market data, and numerical tests. The resulting structure can serve both as a guided curriculum for beginning rates quants and as a reference implementation for researchers and practitioners wishing to inspect calibration, pricing, Greeks, simulation, and validation line by line. Its coverage is complementary to, rather than exhaustive of, the survey: the paper provides the broader model taxonomy and historical synthesis, while the repository supplies executable illustrations for a substantial subset of the sell-side modelling toolkit. The code is intended for education and research and should not be interpreted as investment advice or a production trading system.

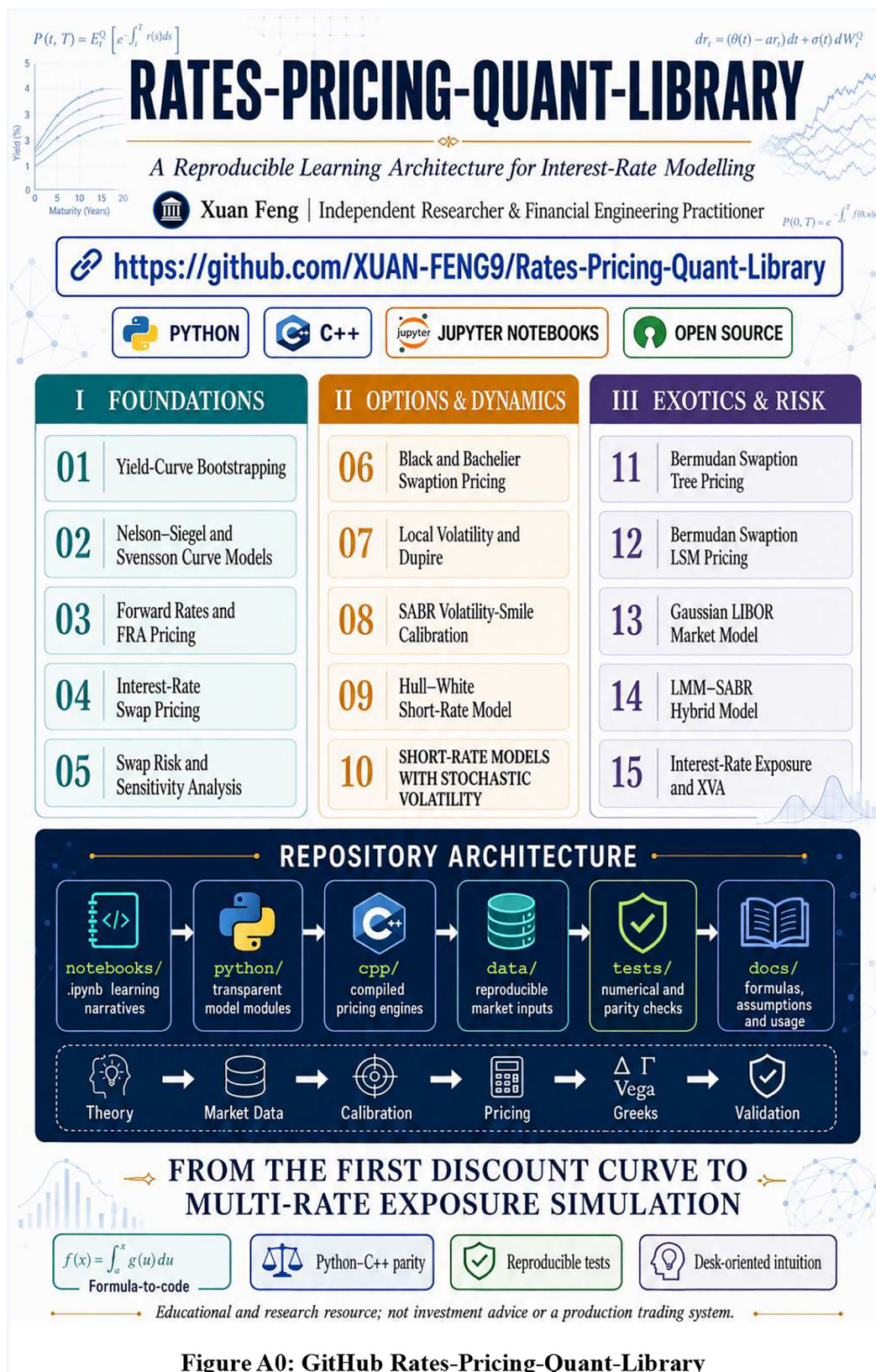


FIGURE A0 | GitHub rates-pricing-quant-library. [Colour figure can be viewed at wileyonlinelibrary.com]